

72nd BPSC PRELIMS

प्रहार

Test Series

Test- 24/08

FULL LENGTH TEST

Solution

English



BPSC TEST-24/8 (SOLUTION)

Q 1. Ans. A

Explanation:

- Koldihwa, located in the Belan Valley of present-day Uttar Pradesh, is regarded as the earliest Neolithic site in the Indian subcontinent to provide evidence of cultivated rice. Excavations have yielded charred rice grains, making it a landmark site for understanding the beginnings of agriculture in India. The site also revealed ground stone tools, handmade pottery, and evidence of domesticated cattle, indicating a settled Neolithic farming community.
- **Additional Facts:**
 - ✓ Mahagara, situated close to Koldihwa in the Belan Valley, is another important Neolithic site known for evidence of cattle domestication, including a cattle pen, but it is not regarded as the earliest site for cultivated rice.
 - ✓ Chopani Mando is a Mesolithic–Neolithic transitional site famous for its hunter-gatherer and early sedentary cultural remains rather than the earliest rice cultivation.
 - ✓ Sohgauna is better known for the Sohgauna Copper Plate, one of the earliest known copper plate inscriptions, and is not a Neolithic agricultural site.

Q 2. Ans. A

Explanation:

- **Statement 1 is correct:** The Aihole Inscription (634 CE), issued during the reign of Pulakeshin II of the Chalukya dynasty, records his victory over Harshavardhana on the banks of the Narmada River. The inscription celebrates Pulakeshin II's military achievements and establishes the Narmada as the boundary between the Chalukya and Harsha's spheres of influence.
- **Statement 2 is incorrect:** The Aihole Inscription was composed by Ravikirti, the court poet of Pulakeshin II, not Banabhatta. Banabhatta was the court poet of Harshavardhana and authored the Harshacharita and Kadambari.
- **Additional Facts:**
 - ✓ The Aihole Inscription is engraved on the Meguti Jain Temple at Aihole in present-day Karnataka.
 - ✓ It is written in Sanskrit using the Kannada (Old Kannada) script.

- ✓ The inscription is an important source for the political history and genealogy of the Early Chalukyas of Badami and is also notable for mentioning the Mahabharata War and the Kali Era for chronological reference.

Q 3. Ans. B

Explanation:

- Kalibangan, located in present-day Rajasthan on the banks of the Ghaggar River, has yielded the earliest archaeologically preserved grid-patterned ploughed field from its pre-Harappan layer. The field displays a criss-cross pattern of furrows running in two directions, indicating mixed cropping, probably of mustard and gram. This discovery provides the clearest and most widely accepted evidence of early agricultural practices in the Harappan cultural tradition. **Kalibangan** is one of the few Harappan sites where **fire altars** have also been discovered.

Q 4. Ans. D

Explanation:

- The Pautavadhyaksha was the superintendent responsible exclusively for regulating weights and measures in the Arthashastra of Kautilya (Chanakya). He ensured the accuracy and standardisation of weights, balances, and measuring instruments, thereby preventing fraud in commercial transactions and maintaining fairness in trade.
- The Arthashastra describes a highly organised administrative system with numerous Adhyakshas (superintendents) overseeing specialised departments.
- Other important officials include the Akaradhyaksha (Superintendent of Mines), Sitadhyaksha (Superintendent of Agriculture), Lavanadhyaksha (Superintendent of Salt), and Suradhyaksha (Superintendent of Liquor).
- **Additional Facts:**
 - ✓ The Panyadhyaksha was the Superintendent of Trade, responsible for regulating the sale, pricing, and distribution of commodities.
 - ✓ The Sulkadhyaksha supervised the collection of customs duties, tolls, and taxes on imports and exports.
 - ✓ The Navadhyaksha was the Superintendent of Shipping, responsible for the management of waterways, ports, boats, and river transport.

Q 5. Ans. B**Explanation:**

- **Statement 1 is correct:** Makkhali Gosala was the founder of the Ajivika sect. According to Jain tradition, he was initially associated with Vardhamana Mahavira for several years before parting ways and establishing the Ajivika order as an independent religious sect.
- **Statement 2 is incorrect:** The Ajivikas did not believe that human effort alone determines destiny. They upheld the doctrine of Niyati (absolute determinism), according to which every event in the universe, including human life and liberation, is predetermined. They rejected the idea that karma or personal effort could alter one's fate.
- **Additional Facts:**
 - ✓ The Ajivika sect was one of the major Śramaṇa traditions alongside Buddhism and Jainism during the 6th century BCE.
 - ✓ Emperor Ashoka and later Dasharatha Maurya donated the famous Barabar and Nagarjuni cave complexes in Bihar to the Ajivikas.
 - ✓ The Ajivikas practised severe asceticism, but unlike Buddhists and Jains, they considered free will ineffective because all events were governed by Niyati (fate).

Q 6. Ans. D**Explanation:**

- **Lichchhavis — Vaishali**
 - ✓ The Lichchhavis were one of the most prominent republican clans of ancient India. Their capital was Vaishali (in present-day Bihar), which also served as the headquarters of the Vajji Confederacy. Vaishali is closely associated with both Buddhism and Jainism and is regarded as one of the world's earliest republican centres.
- **Mallas — Kusinara**
 - ✓ The Mallas were a republican clan with important centres at Kusinara (Kushinagar) and Pava. Kusinara is of immense Buddhist significance as it was here that Gautama Buddha attained Mahaparinirvana.
- **Shakyas — Kapilavastu**
 - ✓ The Shakyas were a republican clan whose capital was Kapilavastu. It was the homeland of Prince Siddhartha Gautama (Buddha) before his renunciation. Kapilavastu is identified with sites on the India–Nepal border, with

Tilaurakot (Nepal) and Piprahwa (Uttar Pradesh) being the principal claimants.

Q 7. Ans. B**Explanation:**

- **Gondophares — Indo-Parthian ruler**
 - ✓ Gondophares was the founder and most prominent ruler of the Indo-Parthian Kingdom, which flourished in north-western India and eastern Iran during the 1st century CE. He is traditionally associated with the visit of St. Thomas the Apostle, though this remains a matter of historical debate.
- **Menander — Indo-Greek king**
 - ✓ Menander (Milinda) was one of the greatest Indo-Greek rulers, who ruled from Sagala (modern Sialkot). He is renowned for the Milinda Panha (Questions of King Milinda), a Buddhist text recording his philosophical dialogue with the monk Nagasena. Buddhist tradition suggests that he embraced Buddhism.
- **Maues — Early Kushana emperor**
 - ✓ This pair is incorrect. Maues (Moga) was the first prominent Indo-Scythian (Saka) ruler in north-western India, not a Kushana emperor. The Kushana dynasty was founded later by Kujula Kadphises, followed by rulers such as Vima Kadphises and Kanishka I.

Q 8. Ans. B**Explanation:**

- **Sabha — Council of tribal elders and elites**
 - ✓ The Sabha was a smaller and more exclusive assembly of the Rig Vedic period, consisting mainly of tribal elders, nobles, and influential persons. It acted as an advisory body to the king and discussed important judicial, administrative, and political matters. Because of its elite composition, it is often regarded as the council of tribal aristocracy.
- **Samiti — General assembly of the entire folk tribe**
 - ✓ The Samiti was the larger and more representative popular assembly comprising the members of the tribe (jana). It discussed matters of public importance such as the election or confirmation of the king, war, peace, and other major tribal affairs. It represented the collective will of the people and was considered more democratic than the Sabha.
- **Vidatha — Oldest folk assembly with secular and religious functions**

- ✓ The Vidatha is regarded as the earliest assembly mentioned in the Rig Veda. It performed religious, military, economic, and social functions. Both men and women are believed to have participated in its proceedings, making it the broadest and oldest popular institution of the early Vedic period.
- **Gana — Small tribal troop or corporate body**
 - ✓ The term Gana originally referred to a tribal group, clan, or corporate body rather than a formal political assembly. In later periods, it came to denote republican states (Gana-Sanghas) governed collectively by clan chiefs instead of hereditary monarchs.
- ✓ After being placed under house arrest at Agra by Aurangzeb, Shivaji executed a daring escape in 1666. According to the widely accepted traditional account, he and his son Sambhaji escaped by hiding in baskets used for distributing sweets and fruits to religious establishments.
- **Battle of Palkhed (1728) — Defeat of the Nizam by Baji Rao I**
 - ✓ The Battle of Palkhed (1728) was fought between Peshwa Baji Rao I and Nizam-ul-Mulk Asaf Jah I. Baji Rao's brilliant cavalry manoeuvres surrounded the Nizam, forcing him to accept the Treaty of Mungi-Paithan. The battle is regarded as a classic example of mobile warfare.

Q 9. Ans. A

Explanation:

- **Statement 1 is correct:** Aurangzeb re-imposed the jizya on non-Muslims in 1679, reversing the policy of religious tolerance followed by Akbar, who had abolished the tax in 1564. The measure became one of the most debated policies of Aurangzeb's reign and reflected his orthodox religious outlook.
- **Statement 2 is correct:** The Fatwai-i-Alamgiri (also known as Fatawa-e-Hindiya) was a comprehensive digest of Hanafi Islamic law compiled under Aurangzeb's patronage. Prepared by a large body of Islamic scholars, it served as an authoritative legal reference for judges and administrators throughout the Mughal Empire.
- **Statement 3 is incorrect:** Aurangzeb did not abolish the office of the Muhtasib. Instead, he revived and strengthened it. The Muhtasib was responsible for enforcing public morality, supervising markets, preventing practices considered un-Islamic, and ensuring adherence to Islamic regulations.

Q 10. Ans. C

Explanation:

- **Treaty of Purandar (1665) — Shivaji surrendered 23 forts to the Mughals**
 - ✓ The Treaty of Purandar (1665) was concluded between Shivaji and the Mughal commander Raja Jai Singh I. Under its terms, Shivaji agreed to surrender 23 forts to the Mughals while retaining 12 forts and accepted Mughal suzerainty over the ceded territories. His son Sambhaji was also to serve the Mughal emperor.
- **Agra Episode (1666) — Shivaji escaped concealed in sweet-meat baskets**

Q 11. Ans. B

Explanation:

- The Third Battle of Panipat (14 January 1761) was fought between the Marathas, led by Sadashivrao Bhau, and the forces of Ahmad Shah Abdali (Ahmad Shah Durrani). One of the key tactical advantages enjoyed by Abdali was the effective use of mobile zamburaks—small swivel guns mounted on camels.
- These camel-mounted guns combined mobility with firepower, allowing Abdali's forces to quickly reposition artillery on the battlefield and provide continuous support to cavalry units. Along with disciplined Afghan cavalry and field artillery, the zamburaks played an important role in disrupting Maratha formations and contributed significantly to the Afghan victory.

Q 12. Ans. A

Explanation:

- The Indian Councils Act of 1892 was regarded by the early nationalists as an inadequate and disappointing constitutional reform because it failed to introduce meaningful representative government or responsible administration.
- **The Assertion (A) is correct:** Although the Act slightly enlarged the Central and Provincial Legislative Councils and introduced an indirect element of election through nominations recommended by certain bodies, it fell far short of the demands of the Indian National Congress for elected majorities and greater legislative powers.
- **The Reason (R) is also correct and correctly explains the assertion:** The Act retained the official

majority in the Central Legislative Council, ensuring continued government control over legislation. While members were permitted to discuss the annual budget and ask questions, they could neither vote on the budget nor move resolutions, leaving public finances firmly under executive control. Thus, Indians were denied any effective authority over administration or financial policy.

Q 13. Ans. B

Explanation:

- The demand for the establishment of agricultural banks was one of the important economic demands of the early nationalists. They argued that Indian peasants were heavily dependent on village moneylenders, who charged exorbitant rates of interest, trapping cultivators in chronic indebtedness. Agricultural banks were proposed to provide cheap institutional credit, enabling peasants to escape the exploitative grip of moneylenders.

Q 14. Ans. B

Explanation:

- The Hindu Marriage Act, 1955, was a landmark post-independence social reform legislation that codified Hindu marriage laws. It made monogamy compulsory for Hindus, rendering a second marriage during the lifetime of a spouse void. The Act also introduced divorce on specified legal grounds, replacing the traditional concept of the indissolubility of Hindu marriage.
- **Additional Facts:**
 - ✓ The Hindu Marriage Act, 1955 formed part of the Hindu Code Bills, which modernised Hindu personal law after Independence.
 - ✓ Along with the Hindu Succession Act, 1956, the Hindu Minority and Guardianship Act, 1956, and the Hindu Adoptions and Maintenance Act, 1956, it significantly improved the legal status of Hindu women by introducing monogamy, divorce, adoption rights, and reforms in inheritance.

Q 15. Ans. D

Explanation:

- **Statement 1 is correct:** Sayyid Ahmad Khan held that the Quran alone was the authoritative religious text in Islam, while other Islamic writings and commentaries were secondary. He argued that these secondary sources should not be treated as infallible.

- **Statement 2 is correct:** He maintained that any interpretation of the Quran must be consistent with human reason, science, and the laws of nature. According to him, if an interpretation conflicted with reason or scientific knowledge, it represented a misinterpretation rather than a defect in the Quran itself.

● **Additional Facts:**

- ✓ Sayyid Ahmad Khan was the founder of the Aligarh Movement and established the Muhammadan Anglo-Oriental College at Aligarh in 1875, which later developed into Aligarh Muslim University.

Q 16. Ans. B

Explanation:

- The Muhammadan Literary Society, founded in 1863 at Calcutta, was one of the earliest organisations to promote modern Western education among Indian Muslims. It encouraged discussions on religious, social, and political issues in the light of modern ideas and sought to persuade the upper and middle-class Muslims to embrace Western education.
- The Muhammadan Literary Society (1863) marked the beginning of organised Muslim social and educational reform in British India. It laid the intellectual foundation for the later Aligarh Movement, which sought to reconcile Islamic thought with modern scientific education and Western learning.

Q 17. Ans. A

Explanation:

- The Anti-Partition Movement against the Partition of Bengal was formally initiated on 7 August 1905, when a massive public meeting was held at the Town Hall in Calcutta. The meeting resolved to oppose the partition and launched the Swadeshi and Boycott Movement. Delegates dispersed from the meeting to spread the movement throughout Bengal.
- **Additional Facts:**
 - ✓ The movement against the Partition of Bengal transformed the Indian national movement by popularising the Swadeshi, Boycott, National Education, and Passive Resistance programmes. The slogan 'Vande Mataram' emerged as the rallying cry of the movement, while Rabindranath Tagore composed Amar Sonar Bangla, which later became the national anthem of Bangladesh.

Q 18. Ans. C

Explanation:

- During the Anti-Partition Movement, Ananda Mohan Bose laid the foundation of the Federation Hall on 16 October 1905, the day the Partition of Bengal came into effect. The Hall was intended to symbolise the indestructible unity of Bengal despite its administrative division by the British. Bose addressed a gathering of over 50,000 people on the occasion.
- 16 October 1905, the day the Partition of Bengal took effect, was observed as a day of national mourning. People observed hartals, fasting, bathed in the Ganga, tied Raksha Bandhan threads to symbolise unity, and raised the slogan “Vande Mataram,” which became the rallying cry of the Swadeshi Movement.

Q 19. Ans. B

Explanation:

- **Statement 1 is correct:** The reforms introduced separate electorates for Muslims for the first time. Under this system, Muslim voters elected only Muslim representatives from separate constituencies. The British justified it as protection for the Muslim minority, but it became a major instrument of the divide-and-rule policy and contributed to the growth of communal politics in India.
- **Statement 2 is incorrect:** The reforms did not establish responsible government. The Executive remained responsible to the British Government, not to the Legislative Councils. The councils possessed only limited **deliberative powers and could neither control the Executive nor remove it from office.**
- **Statement 3 is correct:** The reforms reserved certain elected seats for big landlords and British capitalists. In the Imperial Legislative Council, out of 27 elected members, 6 represented landlords, and 2 represented British commercial interests, reflecting the British policy of protecting loyal and vested interests rather than promoting representative government.

Q 20. Ans. C

Explanation:

- **Statement 1 is correct:** Under martial law, postal communications were subjected to censorship, and the Badshahi Mosque was closed for six weeks.

These measures formed part of the extraordinary restrictions imposed on civil liberties in Punjab.

- **Statement 2 is correct:** The authorities commandeered Indian-owned vehicles for European use and disrupted civic utilities by cutting off electricity and water supplies to Indian households in several **localities. Such actions were among the many punitive measures documented after the massacre.**
- **Additional Facts:**
 - ✓ In protest against these atrocities, Rabindranath Tagore renounced his knighthood in 1919, describing British honours as incompatible with the humiliation inflicted upon the Indian people.

Q 21. Ans. C

Explanation:

- **Statement 1 is correct:** The appointment of the “all-white” Simon Commission caused a direct split within the All-India Muslim League in 1927. The majority faction, led by Muhammad Ali Jinnah, held its annual session in Calcutta and resolved to boycott the Commission at every stage alongside the Indian National Congress. This faction temporarily agreed to the principle of joint electorates (via the Delhi Proposals) to present a united nationalist front.
 - ✓ Conversely, a rival conservative faction led by Sir Muhammad Shafi held a separate session in Lahore. The Shafi faction rejected the joint electorate proposal and chose to fully cooperate with the Simon Commission, arguing that boycotting the British would politically isolate the Muslim community and jeopardise their minority safeguards. Therefore, the boycott was specifically championed by the Jinnah-led faction rather than a unanimous Muslim League.
- **Statement 2 is correct:** The Simon Commission was appointed in November 1927, whereas the Nehru Report, prepared under the chairmanship of Motilal Nehru, was finalised in August 1928. The report was drafted as an Indian alternative to the constitutional proposals expected from the Commission.
- **Additional Facts:**
 - ✓ The Simon Commission was appointed by the British Government in November 1927 to examine constitutional reforms in India. Since all its members were British, its appointment was widely viewed as an insult to Indian aspirations for self-government and was boycotted by most political parties.

Q 22. Ans. C

Explanation:

- At the Lahore Session of the Indian National Congress, the Congress adopted Poorna Swaraj (Complete Independence) as its ultimate objective. It declared 26 January 1930 as the first Independence Day, to be observed annually, with the people taking a solemn pledge to fight for complete freedom from British rule.
- The Lahore Session (December 1929), presided over by Jawaharlal Nehru, adopted Poorna Swaraj as the Congress objective. It also authorised the launch of the Civil Disobedience Movement, the detailed programme of which was later formulated by Mahatma Gandhi, culminating in the Dandi March on 12 March 1930.

Q 23. Ans. C

Explanation:

- **Statement 1 is correct:** Violation of forest laws became an important feature of the movement in Maharashtra, Karnataka, and the Central Provinces. People entered reserved forests, collected forest produce, and grazed cattle in defiance of colonial restrictions.
- **Statement 2 is correct:** The refusal to pay the rural chaukidan (village watchman) tax was a prominent form of protest in eastern India, particularly in parts of Bengal and Bihar, where villagers challenged the colonial administration by withholding this tax.
- **Additional Facts:**
 - ✓ The Civil Disobedience Movement began with Mahatma Gandhi's Dandi March on 12 March 1930. It witnessed mass participation of women, widespread boycott of foreign goods, refusal to pay land revenue in several regions, and the emergence of regional leaders such as Khan Abdul Ghaffar Khan, whose Khudai Khidmatgar mobilised the Pathans of the North-West Frontier Province through non-violent resistance.

Q 24. Ans. A

Explanation:

- During the Civil Disobedience Movement, one of the most remarkable demonstrations of nationalist sentiment within the colonial Indian Army occurred when two platoons of Garhwali soldiers refused to fire on non-violent demonstrators at Peshawar in 1930. Despite knowing that disobedience would

lead to court-martial and imprisonment, they chose not to obey the order, reflecting the growing influence of nationalism within the armed forces.

- The Peshawar episode (1930) occurred alongside the non-violent movement led by Khan Abdul Ghaffar Khan and his Khudai Khidmatgar. The refusal of the Garhwali soldiers to fire on civilians is regarded as one of the earliest and most significant examples of nationalist sentiment within the colonial armed forces, foreshadowing later developments such as the INA movement and the Royal Indian Navy Mutiny of 1946.

Q 25. Ans. D

Explanation:

- According to the reports of the Joint Committee on Indian Constitutional Reform (Session 1933-34), the franchise under the 1935 Act was designed to extend voting rights to roughly 14% of the total population of British India (amounting to approximately 35 million people, up from the roughly 3% franchised under the 1919 Act).
- **The criteria of the voting rights were highly restrictive:**
 - ✓ **Property Qualifications:** Voting rights were not based on adult franchise; they were strictly tied to property ownership, land revenue payment, and tenancy rights.
 - ✓ **Educational Qualifications:** Literacy was a prerequisite for certain categories of voters.
 - ✓ **Gender Disparity:** Although the Act extended the franchise to women (creating separate electorates for them), the ratio of male to female voters remained highly skewed at roughly 4:1 or 5:1 depending on the province, because the qualification was often tied to being the wife of a property owner.

Q 26. Ans. A

Explanation:

- **Pair 1 is correctly matched:** Ambikagiri Roy Chaudhuri (Raichoudhury) was a fiercely patriotic Assamese poet, writer, and freedom fighter. Because of his fiery revolutionary writings and unyielding spirit, he is celebrated as 'Assam Kesari' (The Lion of Assam).
- **Pair 2 is correctly matched:** During the Non-Cooperation Movement (1921), massive strikes broke out among tea plantation workers (famously leading to the Chargola Exodus). While leaders in Bengal and the Surma Valley (like J.M. Sengupta)

supported the striking labourers, the mainstream Congress leadership in the Brahmaputra Valley, led by Nabin Chandra Bardoloi and Tarun Ram Phookan, actively opposed the strikes. They believed the labourers were not yet ready for non-violent Satyagraha and feared that mixing labour unrest with the political movement would invite brutal colonial repression.

- **Pair 3 is incorrectly matched:** The highly influential Hindi nationalist daily Aaj was founded in 1920 in Varanasi by the philanthropist and freedom fighter Shiv Prasad Gupta (with Babu Rao Vishnu Paradkar as its legendary editor). Munshi Premchand, on the other hand, founded the famous literary magazine Hans in 1930 and also edited journals like Jagaran and Madhuri.

- **Additional Facts:**

- ✓ **The Chagola Exodus (1921):** Driven by miserable working conditions and inspired by rumors of “Gandhi Raj,” thousands of tea garden labourers in the Surma Valley (Assam) walked out of the gardens attempting to return to their home provinces. They were met with brutal police firing at Chandpur railway station.
- ✓ **Shiv Prasad Gupta:** Beyond founding Aaj, Shiv Prasad Gupta was a major financier of the national movement and co-founded the Mahatma Gandhi Kashi Vidyapith in Varanasi along with Bhagwan Das.
- ✓ **Ambikagiri’s Institutional Role:** Ambikagiri Roy Chaudhuri also founded the Asom Sangrakshini Sabha (Assam Preservation Society) to protect the cultural and linguistic identity of the Assamese people.

- **Lord Hastings (4. Third Maratha War):** Serving from 1813 to 1823, his aggressive imperial policy led to the Third Anglo-Maratha War (1817–1819), which was sparked by the British campaign against the Pindaris. This war resulted in the complete destruction of the Maratha Confederacy and the abolition of the Peshwaship.

- **Lord William Bentinck (2. Charter Act of 1833):** Serving from 1828 to 1835, his tenure saw the passage of the Charter Act of 1833. This landmark act centralised British administration in India and elevated him from Governor-General of Bengal to the first official Governor-General of India.

- **Additional Facts:**

- ✓ **The Vellore Mutiny Connection:** William Bentinck was actually the Governor of Madras during the Vellore Mutiny of 1806. He was recalled to England for failing to prevent the uprising, only to return decades later as Governor-General.
- ✓ **Charter Act of 1833:** Beyond centralising power, the Charter Act of 1833 ended the East India Company’s status as a commercial body entirely, reducing it to a purely administrative agency of the British Crown. It also introduced the concept of adding a “Law Member” (Lord Macaulay) to the Governor-General’s Executive Council.
- ✓ **The Pindari Link:** The Third Maratha War is historically inseparable from the Pindari War. The Pindaris were irregular horsemen attached to Maratha armies; Lord Hastings’ massive military mobilisation to hunt them down directly threatened the Maratha chiefs, triggering the final conflict.

Q 27. Ans. A

Explanation:

- **Sir George Barlow (3. Vellore Mutiny):** During his short tenure as acting Governor-General, the Vellore Mutiny broke out in 1806. Indian sepoys mutinied against new dress regulations (like the introduction of a new turban and the prohibition of caste marks and beards) introduced by the Commander-in-Chief of Madras, Sir John Cradock.
- **Lord Minto I (1. Treaty of Amritsar):** Serving from 1807 to 1813, he dispatched Charles Metcalfe to Lahore, resulting in the Treaty of Amritsar in 1809 with Maharaja Ranjit Singh. This treaty established the River Sutlej as the boundary between the Sikh Empire and British territory.

Q 28. Ans. D

Explanation:

- **2. Treaty of Bassein (1802):** Signed on December 31, 1802, between the British East India Company and the Maratha Peshwa Baji Rao II after his defeat at the Battle of Poona. The Peshwa accepted the Subsidiary Alliance, which practically signaled the end of Maratha independence.
- **4. Capture of Bharatpur (1826):** After failing to breach its massive mud walls in 1805, the British finally captured the impregnable fortress of Bharatpur in January 1826 under the command of Lord Combermere, defeating the Jat ruler Durjan Sal.
- **1. Annexation of Lower Burma / Pegu (1852):** Occurred during the tenure of Lord Dalhousie at

the conclusion of the Second Anglo-Burmese War. It gave the British complete control over the eastern coastline of the Bay of Bengal.

- **Second White Mutiny (1859):** Following the Government of India Act 1858, the “European Forces” of the East India Company rebelled against their automatic transfer to the British regular army, demanding a bounty or discharge. It was peacefully resolved by offering them a discharge.

- **Additional Facts:**

- ✓ **The Treaty of Bassein aftermath:** This treaty directly triggered the Second Anglo-Maratha War (1803–1805) because the other Maratha chieftains (Scindias and Bhonsles) refused to accept the Peshwa’s surrender of sovereignty to the British.
- ✓ **The First White Mutiny (1766):** This occurred under Robert Clive. European officers in Bengal mutinied because Clive abolished their double bhatta (field allowance).
- ✓ **Dalhousie and Timber:** A major unstated economic reason for the annexation of Lower Burma (Pegu) was British hunger for the rich teak forests of the region, which were desperately needed for shipbuilding.
- ✓ **Lord Canning’s Legacy:** The 1859 White Mutiny was the final nail in the coffin for the Company’s private army. As a result of this unrest, the separate European forces of the East India Company were permanently abolished and amalgamated into the regular British Army.

Q 29. Ans. C

Explanation:

- **Pair 1 is correctly matched:** The 1901 session of the Indian National Congress was held in Calcutta and presided over by **Dinshaw Edulji Wacha**.
- **Pair 2 is correctly matched:** The 1904 session was held in Bombay and presided over by **Sir Henry Cotton**, a sympathetic British civil servant who had long advocated for Indian rights.
- **Pair 3 is correctly matched:** The 1894 session was held in Madras and presided over by **Alfred Webb**, an Irish parliamentarian and a friend of the Indian nationalist cause.
- **Pair 4 is incorrectly matched:** The 1895 session held in Poona was presided over by **Surendranath Banerjea**, not W.C. Bonnerjee. (W.C. Bonnerjee presided over the 1885 Bombay and 1892 Allahabad sessions).

- **Additional Facts:**

- ✓ **The Gandhi Connection (1901 Calcutta):** The 1901 session under D.E. Wacha is highly testable because it marks **Mahatma Gandhi’s very first appearance** on the Congress platform. During a brief visit to India from South Africa, Gandhi attended this session and moved a resolution supporting the struggle of Indians in South Africa.
- ✓ **The 1895 Poona Controversy:** Surendranath Banerjea’s 1895 session was marked by a massive internal clash regarding the **National Social Conference**. Radicals led by Bal Gangadhar Tilak successfully forced the Congress to stop allowing the Social Conference (which focused on social reforms like widow remarriage) to use the INC pandal, arguing that the Congress should focus purely on political goals.
- ✓ **The Foreign Presidents Checklist:** For exams, you must know all five non-Indian presidents of the INC: **George Yule** (1888, Allahabad), **William Wedderburn** (1889, Bombay & 1910, Allahabad), **Alfred Webb** (1894, Madras), **Henry Cotton** (1904, Bombay), and **Annie Besant** (1917, Calcutta).

Q 30. Ans. D

Explanation:

- **A. Madras Courier (3. 1784):** Historically founded in 1784, it was the first English-language newspaper to be published in the Madras Presidency, initiated by Richard Johnston, the government printer.
- **B. India Gazette (1. 1787):** Historically founded in 1787, it was a pro-establishment rival to Hicky’s Bengal Gazette and the second newspaper printed in India.
- **C. Digdarshana (2. 1818):** Founded in April 1818, this was the first periodical of India published in an Indian language (Bengali). It was started by the Serampore missionaries (specifically John Clark Marshman) to spread knowledge among the youth.
- **D. Bombay Herald (4. 1789):** Founded in 1789, it was the first English newspaper published in the Bombay Presidency.
- **Additional Facts:**
 - ✓ **The First of the First:** James Augustus Hicky founded the Bengal Gazette (also known as the Calcutta General Advertiser) in 1780. It was fiercely critical of Governor-General Warren Hastings, leading to its seizure in 1882.
 - ✓ **The Serampore Trio’s Monopoly:** The

Serampore missionaries (Carey, Marshman, and Ward) didn't just stop at Digdarshana. A month later, in May 1818, they launched Samachar Darpan, which became the first Indian-language (Bengali) newspaper of daily/weekly news, completely transforming vernacular journalism in India.

- ✓ **The Press Liberator:** Sir Charles Metcalfe (Governor-General 1835–1836) is known as the “Liberator of the Indian Press” for repealing the draconian 1823 licensing regulations introduced by John Adams.

Q 31. Ans. C

Explanation:

- **Pair 1 is correct:** The 15th BRICS Summit was held in Johannesburg, South Africa, from 22 to 24 August 2023, under the theme “BRICS and Africa: Partnership for Mutually Accelerated Growth, Sustainable Development and Inclusive Multilateralism.” The Johannesburg II Declaration was adopted, and this summit approved the historic BRICS expansion.
- **Pair 2 is correct:** The 16th BRICS Summit was held in Kazan, Russia, from 22 to 24 October 2024, under the theme “Strengthening Multilateralism for Fair Global Development and Security.” This was the first summit to feature Egypt, Ethiopia, Iran, and the UAE as new full members.
- **Pair 3 is incorrect:** The 17th BRICS Summit was held in Rio de Janeiro, Brazil, on 6-7 July 2025, not in Beijing, China. It was held under the theme “Strengthening Global South Cooperation for More Inclusive and Sustainable Governance,” and Indonesia participated as a new member for the first time.
- **Pair 4 is correct:** The 18th BRICS Summit is being hosted by India in 2026, with New Delhi as the likely venue. India assumed the BRICS chairship on 1 January 2026, succeeding Brazil.

Q 32. Ans. A

Explanation:

- The India-UAE bilateral maritime exercise, earlier known as “Zayed Talwar,” has been rechristened as “Gulf Waves.” The rebranding was formally reflected in official communications, including the Embassy of India in Abu Dhabi.
- The inaugural edition of Zayed Talwar was held in March 2018, with subsequent editions in 2021, 2023, and 2025 before the rebranding.

- The exercise aims to enhance interoperability and synergy between the Indian Navy and the UAE Navy through cross-training on tactics, techniques, and procedures.

Q 33. Ans. A

Explanation:

- C.P. Radhakrishnan was sworn in as the 15th Vice President of India on 12 September 2025 by President Droupadi Murmu at Rashtrapati Bhavan. He assumed office following the resignation of Jagdeep Dhankhar on 21 July 2025, which had triggered the mid-term Vice-Presidential election.
- In the election held on 9 September 2025, Radhakrishnan, the NDA candidate, defeated the joint Opposition candidate former Supreme Court Justice B. Sudershan Reddy by a decisive margin, securing 452 votes to Reddy's 300.
- Before becoming Vice President, C.P. Radhakrishnan served as the Governor of Maharashtra (from July 2024) with additional charge of the Governor of Jharkhand. He previously served as a two-term Member of Parliament from Coimbatore (1998 and 1999) and as the state BJP president of Tamil Nadu.

Q 34. Ans. B

Explanation:

- **Statement 1 is correct:** VB-G RAM G stands for Viksit Bharat–Guarantee for Rozgar and Ajeevika Mission (Gramin). The Bill was introduced in the Lok Sabha on 16 December 2025 by Minister of Rural Development Shivraj Singh Chouhan, passed by the Lok Sabha on 18 December 2025 and by the Rajya Sabha on 19 December 2025, and received Presidential assent from Droupadi Murmu on 21 December 2025. It replaces the Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA), 2005.
- **Statement 2 is incorrect:** The Act increases the statutory employment guarantee from 100 days to 125 days per rural household in a financial year, not 150 days. This is one of the key changes over MGNREGA.
- **Statement 3 is incorrect:** Unlike MGNREGA (where the Centre bore the entire wage cost and most of the material cost), VB-G RAM G is implemented as a Centrally Sponsored Scheme with a 60:40 Centre-State fund-sharing ratio for most states, 90:10 for North-Eastern and Himalayan states, and 100% central funding for Union Territories without legislatures.

- **Statement 4 is incorrect:** The Act allows state governments to notify a period of up to 60 days in a financial year during which works under the scheme will not be undertaken, covering peak sowing and harvesting seasons. The remaining 305 days are available to fulfil the 125-day employment guarantee.

Q 35. Ans. C

Explanation:

- C. Joseph Vijay launched his party, the Tamilaga Vettri Kazhagam (TVK), on February 2, 2024. In the 2026 Tamil Nadu Legislative Assembly elections, his party made a historic electoral debut, emerging as the single-largest party by winning 108 of the 234 assembly seats.
- Following the election results in May 2026, TVK formed a coalition government with the support of several allied parties to reach the majority mark. C. Joseph Vijay was subsequently sworn in as the 22nd Chief Minister of Tamil Nadu on May 10, 2026.

Q 36. Ans. C

Explanation:

- Launched following approval in July 2025, the scheme is a flagship initiative designed to transform India's agricultural sector. It focuses on enhancing agricultural productivity, promoting crop diversification, improving irrigation, and ensuring better access to credit.
- **Selection Criteria:** The 100 districts were identified based on specific parameters, including low agricultural productivity, moderate crop intensity, and below-average credit disbursement.
- **Implementation:** The scheme follows a saturation-based convergence model, integrating 36 existing central schemes from 11 different ministries to provide a coordinated approach to agricultural development.
- It has an annual outlay of ₹24,000 crore, spanning a period of six years (FY 2025–26 to FY 2030–31).

Q 37. Ans. A

Explanation:

Ministry-wise allocation facts for enrichment:

Ministry	BE 2026-27
Ministry of Finance (incl. interest, subsidies)	₹19,72,509 crore (highest overall)
Ministry of Defence	₹7,84,678 crore

Ministry of Road Transport and Highways	₹3,09,875 crore
Ministry of Railways	₹2,81,377 crore
Ministry of Home Affairs	₹2,55,233 crore
Ministry of Rural Development	₹2,73,000 crore (approx.)
Ministry of Education	₹1,39,289 crore
Ministry of Health and Family Welfare	₹1,06,530 crore
Ministry of Housing and Urban Affairs	₹85,522 crore

Q 38. Ans. A

Explanation:

- **Australian Open:** Jannik Sinner successfully defended his men's title by defeating Alexander Zverev, and Madison Keys captured the women's title by defeating Aryna Sabalenka.
- **French Open:** While Carlos Alcaraz did win the men's singles, it was Coco Gauff who won her first Roland Garros women's title by defeating Aryna Sabalenka.
- **Wimbledon:** Iga Swiatek won the women's title, but the men's champion was Jannik Sinner, who defeated Carlos Alcaraz in a four-set final.
- **US Open:** Aryna Sabalenka successfully defended her women's title, but the men's champion was Carlos Alcaraz, who defeated Jannik Sinner.
- **Other Major 2025 Tennis Tournaments & Winners:**
 - ✓ **Nitto ATP Finals (Men's Singles):** Won by Jannik Sinner. Playing in Turin, Italy, Sinner defeated Carlos Alcaraz.
 - ✓ **WTA Finals (Women's Singles):** Won by Elena Rybakina. She defeated Aryna Sabalenka in straight sets.
 - ✓ **International Team Competitions:**
 - **Davis Cup (Men's Team):** Won by Italy. The Italian men secured a historic "threepeat" (their third consecutive title) by defeating Spain 2-0 in the final in Bologna.
 - **Billie Jean King Cup (Women's Team):** Won by Italy. The Italian women also had a dominant year, successfully defending their team title by defeating the United States 2-0 in the final in Shenzhen, China.

Q 39. Ans. D

Explanation:

- **Physiology or Medicine:** Brunkow, Ramsdell & Sakaguchi were awarded for discovering peripheral

immune tolerance through regulatory T-cells (Tregs) and the FOXP3 gene, which prevents the immune system from attacking the body's own tissues.

- **Physics:** Clarke, Devoret & Martinis were awarded for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit, laying the foundation for modern quantum computing.
- **Chemistry:** Kitagawa, Robson & Yaghi were awarded for developing metal-organic frameworks (MOFs), porous crystalline materials used to capture CO₂, harvest water from desert air, and store toxic gases.
- **Literature:** László Krasznahorkai (Hungary) was awarded "for his compelling and visionary oeuvre that, in the midst of apocalyptic terror, reaffirms the power of art."
- **Peace:** María Corina Machado (Venezuela) for promoting democratic rights and peaceful transition from dictatorship in Venezuela.
- **Economics:** Joel Mokyr, Philippe Aghion & Peter Howitt for explaining innovation-driven economic growth through technological progress and the theory of "creative destruction."

Q 40. Ans. B

Explanation:

- **Statement 1 is incorrect:** The 77th Republic Day 2026 had two Chief Guests: António Costa, President of the European Council, and Ursula von der Leyen, President of the European Commission, marking the first time the EU's two top leaders jointly attended India's Republic Day celebrations. Indonesian President Prabowo Subianto was the chief guest of the 76th Republic Day 2025.
- **Statement 2 is correct:** The theme of Republic Day 2026 was "150 Years of Vande Mataram," commemorating the 150th anniversary of Bankim Chandra Chatterjee's composition of the iconic national song in 1876.
- **Statement 3 is correct:** Maharashtra's Tableau won the Best Tableau award in the State/UT Category at the 77th Republic Day Parade 2026.
- **Statement 4 is incorrect:** The Best Tableau in the Ministries/Departments Category was won by the Ministry of Culture, not the Ministry of Home Affairs.

Q 41. Ans. A

Explanation:

- **Merriam-Webster:** "Slop" was chosen to describe the flood of low-quality, mass-produced

digital content—often generated by artificial intelligence—that saturates the internet. This perfectly matches the first pair.

- **Oxford:** "Rage bait" was selected to describe online content deliberately designed to elicit anger or outrage in order to drive engagement, clicks, and visibility. This perfectly matches the second pair.
- **Cambridge:** The dictionary actually chose "Parasocial" as its word of the year to highlight the surge in one-sided emotional relationships people form with influencers, celebrities, or even AI companions (the question incorrectly paired it with vibe coding).
- **Collins:** The actual word chosen was "Vibe coding," which is a slang term for generating computer code by giving natural-language prompts to artificial intelligence rather than writing the instructions manually (the question incorrectly paired it with parasocial).
- **Other Notable 2025 Words of the Year**
 - ✓ **Dictionary.com:** Choose "67" (often written out as "six seven"). Spoken aloud, it functions in internet culture as a deliberately vague, noncommittal response, neither a yes nor a no.
 - ✓ **American Dialect Society:** Also selected "Slop," noting that the term was so culturally dominant in 2025 that people dropped the original longer phrase, "AI slop," completely.
 - ✓ **Macquarie Dictionary:** Australia's national dictionary also crowned "Slop" as its word of the year, cementing the term's global reach in 2025.

Q 42. Ans. B

Explanation:

- **Pair 1 is correct:** The Meri Rasoī Yojana was launched by the Punjab government in 2026. It is a household nutrition support initiative designed to provide free quarterly grocery kits (containing pulses, sugar, iodised salt, turmeric, and mustard oil) to approximately 40 lakh economically weaker families, supplementing the standard wheat distribution under the National Food Security Act.
- **Pair 2 is correct:** The West Bengal government launched the Annapurna Yojana (officially rolled out as the Annapurna Bhandar Scheme) in 2026 to replace the older Lakshmir Bhandar scheme. It provides a flat financial assistance of ₹3,000 per month via Direct Benefit Transfer (DBT) to eligible women aged 25 to 60.
- **Pair 3 is incorrect:** The recent Aftercare Scheme

was launched by the Delhi government in 2026, not Maharashtra. The scheme operates under the Juvenile Justice Act to support vulnerable youth who are required to leave Child Care Institutions (CCIs) upon turning 18. It provides a monthly stipend, funding for higher education, and vocational training to ensure they do not face extreme poverty or homelessness as they transition into adulthood.

Q 43. Ans. C

Explanation:

- Project Zero is an anti-corruption drive launched by the Kerala Government (specifically the Vigilance and Anti-Corruption Bureau) in 2026. The initiative aims to completely eliminate bribery and digital graft in government offices. It relies heavily on public participation, offering a ₹5,000 reward for citizens who upload verified video evidence of officials demanding bribes, while keeping the informant's identity confidential.

Q 44. Ans. A

Explanation:

- **Nagori (Nagauri) Ashwagandha → Rajasthan:** Nagauri Ashwagandha is a premium variety of the medicinal plant *Withania somnifera* (Indian ginseng/Indian winter cherry) grown in the arid and semi-arid Nagaur district of Rajasthan. The dry climate and sandy soil of the Marwar region give it longer, thicker roots rich in medicinal alkaloids and berries with a distinctive bright red colour, making it the finest Ashwagandha variety in arid regions. It became Rajasthan's second major agricultural GI product after Sojat Mehendi.
- **Ponduru Khadi → Andhra Pradesh:** Ponduru Khadi is a traditional handspun and handwoven cotton fabric from Ponduru village in Srikakulam district, locally known as Patnulu. It is made from indigenous hill cotton, punasa cotton, and red cotton, with the entire cleaning, spinning, and weaving done manually. Two features make it globally unique: the cotton is cleaned using the jawbone of the Valuga fish, and spinners use single-spindle charkhas with 24 spokes (the Gandhi Charkha). It has an exceptionally high yarn count of 100 to 120, and the GI registration was granted in favour of the Khadi and Village Industries Commission (KVIC).
- **Thuyamalli (Thooyamalli) Rice → Tamil Nadu:** Thooyamalli is a traditional aromatic rice variety of Tamil Nadu, prized for its fine texture and fragrance. It was among five Tamil Nadu products

granted GI status in December 2025 (alongside the Woraiyur cotton sari, Kavindapadi naatu sakkarai, and others), which raised Tamil Nadu's total to 74 GI products.

- **Ozzo Feni → Goa:** Ozzo Feni, produced by Cazcar Distillery in Nanora village near Panaji, became the first feni in India to receive a GI tag (June 2025). Feni is a traditional Goan spirit made by fermenting and double-distilling cashew apples or coconut sap, introduced by Portuguese missionaries in the 1600s. Goa had earlier declared feni its "Heritage Spirit" in 2016. With the GI tag, feni now joins the ranks of region-specific spirits like Scotch and Champagne.
- **Other notable GI tags :**
 - ✓ **Bihar:** Pidiya Painting (Bhojpur), Bawan Buti Saree (Nalanda), and Pattharkatti Stone Craft (Gaya).
 - ✓ Sojat Mehendi (Rajasthan)
 - ✓ Kolkatti/Kolkata Jewellery and Begampur cotton saree (West Bengal)
 - ✓ Hilario Mango and Taleigao brinjal (Goa)
 - ✓ Woraiyur cotton sari and Kavindapadi naatu sakkarai (Tamil Nadu)
 - ✓ Nagauri Pan (also from Rajasthan's Nagaur region).
- GI tags in India are administered under the Geographical Indications of Goods (Registration and Protection) Act, 1999, which came into force on 15 September 2003.
- Darjeeling Tea (West Bengal) was the first Indian product to receive a GI tag in 2004.
- A GI registration is valid for 10 years and is renewable indefinitely.
- GI tags confer collective rights on all producers within the registered geographical region (unlike a patent, which grants exclusive rights to an individual inventor).
- The GI Registry head office is located in Chennai.

Q 45. 45. Ans. A

Explanation:

- **Statement 1 is correct:** During Prime Minister Modi's visit to **Malaysia** (February 2026), India announced Thiruvalluvar scholarships (alongside a Thiruvalluvar Centre at Universiti Malaya) to enable Malaysian students to study in India and to promote the teachings of the Tamil poet-philosopher Thiruvalluvar, strengthening academic and cultural exchange between India and Malaysia.
- **Statement 2 is correct:** During PM Modi's state visit to **Israel** (25-26 February 2026), India and

Israel officially elevated their bilateral relationship to a “Special Strategic Partnership for Peace, Innovation and Prosperity,” with 17 agreements and 27 outcomes across defence, AI, cybersecurity, agriculture, trade, and labour mobility.

- **Statement 3 is incorrect:** India and Israel signed a Memorandum of Understanding (MoU) to officially link India’s Unified Payments Interface (UPI) with Israel’s Money Automated Clearing House (MASAV) for seamless cross-border remittances and payments.
- **Statement 4 is incorrect:** The Fast Patrol Vessel PS Lespwar was gifted by India to **Seychelles** (not Mauritius), during PM Modi’s visit to Seychelles for its Golden Jubilee/50th National Day celebrations (late June 2026). Built by Goa Shipyard Limited, the vessel is meant to strengthen Seychelles’ maritime surveillance and protect its Exclusive Economic Zone. “Lespwar” means “hope” in Seychellois Creole.

Q 46. Ans. A

Explanation:

- **Pair 1 is correct:** On May 1, 2026, Sikkim was officially declared as the first fully paperless state judiciary in the country.
- **Pair 2 is incorrect:** Kerala became India’s first state to establish a dedicated cabinet-level Artificial Intelligence (AI) portfolio in May 2026 to modernise governance and technology-led public policy.
- **Pair 3 is incorrect:** Hazaribag in Jharkhand was declared India’s first pearl farming cluster under the Pradhan Mantri Matsya Sampada Yojana (PMMSY) to promote sustainable rural livelihoods.
- **Pair 4 is incorrect:** Chandigarh, not New Delhi, became India’s first slum-free city after successfully reclaiming land and demolishing its last remaining slum colony.

Q 47. Ans. D

Explanation:

- **Pair 1 is correct:** The Mobile World Congress (MWC) 2026, the world’s premier connectivity, telecommunications, and mobile technology event, took place in Barcelona, Spain.
- **Pair 2 is correct:** The World Hydrogen Summit & Exhibition 2026 was hosted at the Rotterdam Ahoy in Rotterdam, Netherlands, from May 19 to 21, 2026. It is a leading global event focusing on the international hydrogen sector, clean energy solutions, and global decarbonization strategies.

- **Pair 3 is correct:** The Global Energy Leaders’ Summit (GELS) 2025 was held in Puri, Odisha, from December 5 to 7, 2025. The summit was hosted by the Government of Odisha in association with the Tony Blair Institute for Global Change. It is an important event for driving policy regarding India’s Net-Zero 2070 commitments and energy transition, highlighting Odisha’s status as the first Indian state to pioneer comprehensive power sector reforms in the 1990s.
- **Pair 4 is correct:** The 47th ASEAN Summit and Related Meetings officially took place in Kuala Lumpur, Malaysia, in October 2025. The summit marked the culmination of Malaysia’s 2025 ASEAN Chairmanship, which operated under the theme “Inclusivity and Sustainability”. ASEAN leaders formally signed the accession instrument to welcome Timor-Leste as ASEAN’s 11th Member State. Following this summit, Malaysia handed over the ASEAN Chairmanship to the Philippines for the year 2026.

Q 48. Ans. B

Explanation:

- **Statement 1 is correct:** The EPI is released by NITI Aayog and is aligned with India’s target of USD 1 trillion in merchandise exports by 2030 and the vision of Viksit Bharat @2047.
- **Statement 2 is correct:** Of the four pillars, Business Ecosystem carries the highest weightage at 40%. The remaining three, namely Export Infrastructure, Policy and Governance, and Export Performance, carry 20% each.
- **Statement 3 is incorrect:** Maharashtra topped the Large States category, followed by Tamil Nadu and Gujarat.
- **Statement 4 is correct:** Uttarakhand topped the combined Small States, North Eastern States and Union Territories category, followed by Jammu and Kashmir.
- **Additional facts:**
 - ✓ **Releasing body and edition:** Released by NITI Aayog on 14 January 2026. This is the 4th edition of the EPI. The first edition was published in August 2020.
 - ✓ **Framework:** Structured around 4 pillars, 13 sub-pillars, and 70 indicators.
 - ✓ **Pillars and weightages:**
 - **Export Infrastructure:** 20 percent (sub-pillars: Utilities, Logistics)
 - **Business Ecosystem:** 40 percent

(sub-pillars: Macroeconomy, Cost competitiveness, Human capital, Finance and credit accessibility, MSME ecosystem, Industrial and innovation environment)

- **Policy and Governance:** 20 percent (sub-pillars: State-level policy support and governance, Regulatory environment and compliance)
- **Export Performance:** 20 percent (sub-pillars: State level exports and trends, Export promotion and facilitation, Export portfolio and market access)
- ✓ **New dimensions added in the 2024 edition:** macroeconomic stability, cost competitiveness, human capital, financial access, and the MSME ecosystem.
- ✓ **Classification:** States and UTs are grouped into Large States, Small States, North East States, and Union Territories. Within each group, they are ranked as Leaders, Challengers, and Aspirers.
- ✓ **Top 5 Large States:** Maharashtra, Tamil Nadu, Gujarat, Uttar Pradesh, Andhra Pradesh.
- ✓ **Top 5 Small States, NE States and UTs:** Uttarakhand, Jammu and Kashmir, Nagaland, Dadra and Nagar Haveli & Daman and Diu, and Goa.
- ✓ **Other pointers:** Districts have been emphasised as the core units of export competitiveness.

Q 49. Ans. C

Explanation:

- **Indore** secured 1st rank in **Category-1** with a perfect score of 200 out of 200 and received a cash award of Rs. 1.5 crore. It planted over 16 lakh trees in the last year, earning a Guinness World Record, and runs public transport through 120 electric buses and 150 CNG buses.
- **Other rankings in Category-1 (population over 10 lakh):** Jabalpur secured 2nd rank with a score of 199 out of 200 and received Rs. 1 crore; it set up an 11 MW waste-to-energy plant. **Agra** and **Surat** jointly secured 3rd rank with a score of 196 out of 200, each receiving Rs. 25 lakh. Agra remediated a legacy waste dump site and carried out Miyawaki plantation, while Surat implemented an EV policy offering incentives and tax benefits and maintains 38% green cover.
- **Category-2 (population between 3 and 10 lakh):** **Amravati** ranked 1st with a perfect score of 200 out of 200 (cash award Rs. 75 lakh), improving 340 km

of road infrastructure and converting 19 acres of barren land into dense forest. **Jhansi** and **Moradabad** jointly ranked 2nd with 198.5 out of 200 (Rs. 25 lakh each); Jhansi developed Miyawaki forests while Moradabad focused on road infrastructure and construction and demolition waste management. **Alwar** ranked 3rd with 197.6 out of 200 (Rs. 25 lakh), remediating its legacy waste dump.

- **Category-3 (population under 3 lakh):** **Dewas** ranked 1st with 193 out of 200 (Rs. 37.50 lakh), shifting industries to cleaner fuels. **Parwanoo** ranked 2nd with 191.5 out of 200 (Rs. 25 lakh), focusing on end-to-end road pavement. **Angul** ranked 3rd with 191 out of 200 (Rs. 12.50 lakh), working on road infrastructure and public outreach.

Q 50. Ans. B

Explanation:

- The State Energy Efficiency Index (SEEI) is collaboratively developed and published by the **Bureau of Energy Efficiency (BEE)**, a statutory body under the Ministry of Power, in association with the **Alliance for an Energy Efficient Economy (AEEE)**. The SEEI's primary purpose is to benchmark, track, and compare the implementation of energy efficiency initiatives across Indian States and Union Territories.
- **Additional Facts:**
 - ✓ The latest index assesses 36 States and UTs using 66 indicators spread across seven key demand sectors: Buildings, Industry, Municipal Services, Transport, Agriculture, DISCOMs, and Cross-Sector initiatives.
 - ✓ **Performance Categories:** States are classified into four distinct categories based on their total evaluation score:
 - **Front Runners:** >60% score
 - **Achievers:** 50–60% score
 - **Contenders:** 30–50% score
 - **Aspirants:** <30% score
 - ✓ **Top Performers (By Energy Consumption Group):** To ensure a fair comparison, states are grouped by their Total Final Energy Consumption (TFEC). The top states in each group are:
 - ✓ **Group 1 (>15 MToE):** Maharashtra
 - ✓ **Group 2 (5–15 MToE):** Andhra Pradesh
 - ✓ **Group 3 (1–5 MToE):** Assam
 - ✓ **Group 4 (<1 MToE):** Tripura
 - Bihar is in Group 2 at 6th position.
- **Sectoral Milestones:**

- ✓ 24 states have successfully notified the Energy Conservation Building Code (ECBC) 2017.
- ✓ 31 states have adopted dedicated electric mobility (EV) policies.
- ✓ Kerala emerged as a standout in the agriculture sector, achieving a massive 74% adoption rate for energy-efficient or solar-powered agricultural pumps.
- ✓ All 36 States and UTs have now successfully developed State Energy Efficiency Action Plans (SEEAPs).

- Zinc
- Iron
- Lead
- Hydrogen
- Copper
- Silver
- Gold

Q 51. Ans. B

Explanation:

- The modern periodic table is arranged based on the **atomic number** of elements. Elements placed in the **same vertical column (group)** have the **same number of valence electrons** in their outermost shell. Since chemical properties depend primarily on the valence electrons, elements in the same group exhibit **similar chemical properties**.
- Examples:
 - ✓ **Group 1 (Alkali metals):** Li, Na, K
 - All have 1 valence electron.
 - ✓ **Group 17 (Halogens):** F, Cl, Br
 - All have 7 valence electrons.
 - ✓ **Group 18 (Noble gases):** Ne, Ar, Kr
 - Their outermost shell is completely filled (8 electrons, except He which has 2).
- Thus, the common characteristic of elements in the same group is their **number of valence electrons**.

Q 52. Ans. C

Explanation:

- **Assertion (A) is true :** metals **high in the reactivity series** (such as potassium, sodium, calcium, magnesium, and aluminium) **cannot be extracted by reducing their oxides with carbon** because they have a greater affinity for oxygen than carbon. Instead, these metals are extracted by the **electrolytic reduction** (electrolysis) of their molten compounds. Since **aluminium is high in the reactivity series**, it is extracted by **electrolysis**.
- **Reason (R) is False:** aluminium **high in the reactivity series**, above zinc, iron, and lead.
 - ✓ Reactivity series
 - Potassium
 - Sodium
 - Calcium
 - Magnesium
 - Aluminium

REACTIVITY SERIES (OR ACTIVITY SERIES) OF METALS		
	Potassium K	Most reactive metal
	Sodium Na	
	Barium Ba	
	Calcium Ca	
	Magnesium Mg	
	Aluminium Al	
	Zinc Zn	
	Iron Fe	
	Nickel Ni	
	Tin Sn	
	Lead Pb	
	Hydrogen (H)	Decreasing chemical reactivity
	Copper Cu	
	Mercury Hg	
	Silver Ag	
	Gold Au	
	Platinum Pt	
		Least reactive metal

- ✓ Thus, aluminium **does not lie low** in the reactivity series.

Q 53. Ans. A

Explanation:

- **Statement 1 is correct:** catenation is the property by which atoms of an element form bonds with other atoms of the same element. Carbon exhibits this property to a much greater extent than silicon because carbon atoms are smaller in size and form very strong carbon-carbon (C–C) covalent bonds. These strong bonds allow carbon atoms to combine into long chains, branched chains, and ring structures, giving rise to an enormous variety of stable compounds. Therefore, catenation is more pronounced in carbon than in silicon.
- **Statement 2 is correct:** carbon forms very strong covalent single bonds with other carbon atoms due to its small atomic size. The strength and stability of the carbon-carbon (C–C) bond enable carbon atoms to remain linked together in long chains and complex structures without breaking easily. This strong self-linking ability is the fundamental reason for carbon's exceptional catenation and the existence of millions of carbon compounds. Hence,

the statement that carbon forms strong single bonds with itself is correct.

- **Statement 3 is incorrect:** the vast number of carbon compounds is a direct consequence of **both catenation and tetravalency**. Carbon has four valence electrons and therefore forms four stable covalent bonds with carbon as well as with elements such as hydrogen, oxygen, nitrogen, and halogens. This tetravalency allows carbon to produce an enormous variety of compounds with different structures and properties. Thus, the ability of carbon to form numerous compounds is closely related to its tetravalency, **making this statement incorrect**.

Q 54. Ans. C

Explanation:

- **Statement 1 is correct:** hydrogenation is the process of adding hydrogen to **unsaturated hydrocarbons or vegetable oils** in the presence of a catalyst such as **nickel (Ni)**. During this process, hydrogen is added across the carbon-carbon double or triple bonds, converting the unsaturated oil into a saturated product.
- **Statement 2 is incorrect:** hydrogenation is an **addition reaction**, not a decomposition reaction. In an addition reaction, atoms or groups of atoms are added to an unsaturated compound, resulting in the formation of a single saturated product. During hydrogenation, hydrogen molecules add across the double or triple bonds of unsaturated compounds without breaking the compound into simpler substances.
- **Statement 3 is correct:** unsaturated compounds contain one or more **double or triple carbon-carbon bonds**. During hydrogenation, hydrogen is added across these multiple bonds, converting them into **single bonds**. As a result, the unsaturated compound becomes a **saturated compound**. This principle is applied industrially to convert liquid vegetable oils into solid or semi-solid fats such as vanaspati. Therefore, the statement is correct.

Q 55. Ans. D

Explanation:

- **Statement 1 is correct:** compounds that have the **same molecular formula but different structural arrangements of atoms** are called **isomers**, and the phenomenon is known as **isomerism**. Although isomers contain the same kinds and numbers of atoms, the atoms are arranged differently, resulting

in compounds with different structural formulas and, often, different physical and chemical properties.

- **Statement 2 is correct: isomerism becomes possible from compounds containing four or more carbon atoms** and becomes increasingly common as the number of carbon atoms in the molecule increases. Larger carbon chains can be arranged in multiple ways while retaining the same molecular formula, giving rise to a greater number of structural isomers. Hence, structural isomerism is more commonly observed in carbon compounds with longer carbon chains.

Q 56. Ans. C

Explanation:

- **Statement 1 is correct:** work done by a force depends on the direction of the force with respect to the displacement of the object. When the force acts in the direction opposite to the displacement, the work done by the force is **negative**. NCERT explains this using the relation $Fs \cos \theta$, where θ is the angle between force and displacement, where the angle between the force and displacement determines the sign of the work. If the angle is **180°**, the force opposes the motion and the work done is negative. A common example is the **frictional force**, which acts opposite to the direction of motion and therefore performs negative work. Hence, the statement is correct.
- **Statement 2 is correct:** work is done only when the applied force has a component in the direction of displacement. If the force acts **perpendicular (at 90°)** to the direction of displacement, it has no component along the displacement. Consequently, the work done is **zero**. For example, when a person walks on a horizontal surface carrying a load on the head, the upward force exerted on the load is perpendicular to the horizontal displacement. Therefore, no work is done by this upward force on the load. Hence, the statement is correct.

Q 57. Ans. C

Explanation:

- Assertion (A) is true Quicklime (Calcium oxide, CaO) reacts vigorously with water to produce slaked lime (Calcium hydroxide, Ca(OH)_2). This reaction is vigorous and a large amount of heat is evolved.
- Reason (R) is False: A reaction that releases heat is called an **exothermic reaction**, not an endothermic reaction.

- ✓ **Exothermic reaction:** Heat is evolved.
- ✓ **Endothermic reaction:** Heat is absorbed.
- ✓ Since heat is produced during the reaction between quicklime and water, it is **exothermic**. Therefore, the reason is **false**.

Q 58. Ans. B

Explanation:

- The speed of light depends on the optical density of the medium through which it travels. Light travels **fastest in a vacuum** and slower in material media such as air, water, and glass.
- When light passes **from a rarer medium (such as air) to a denser medium (such as water or glass)**, its **speed decreases**. This decrease in speed causes the light ray to bend towards the normal, a phenomenon known as **refraction**. Conversely, when light passes from a denser medium to a rarer medium, its speed increases and the ray bends away from the normal. Therefore, the speed of light decreases when it enters a **denser medium from a rarer medium**.

Q 59. Ans. B

Explanation:

- **Radioactivity** is the **spontaneous disintegration (decay) of an unstable atomic nucleus**, accompanied by the emission of **alpha (α), beta (β), or gamma (γ) radiations**. During radioactive decay, changes occur in the **nucleus** of the atom, such as a change in the number of protons and/or neutrons, which may transform one element into another. The electrons outside the nucleus are not responsible for radioactivity, as the process is purely a **nuclear phenomenon**. Therefore, radioactivity is a process involving a change in the **nucleus** of an atom.

Q 60. Ans. A

Explanation:

- **Assertion (A) is true:** a **galvanic (voltaic) cell** is an electrochemical cell in which a **spontaneous redox reaction** produces electrical energy.
 - ✓ **In a galvanic cell:**
 - Oxidation occurs at the anode.
 - Reduction occurs at the cathode.
 - Electrons released during oxidation travel through the external circuit, generating an electric current.
 - ✓ Thus, **chemical energy is converted into electrical energy**.
- **Reason (R) is true:** galvanic cells operate because

the redox reaction proceeds **spontaneously**. Since the reaction is spontaneous, electrons flow naturally from the anode to the cathode through the external circuit, producing electrical energy without requiring an external power source. Therefore, the reason is **true**.

Q 61. Ans. D

Explanation:

- **Pair 1 is correct:** the **Right-hand Thumb Rule** is used to determine the direction of the magnetic field produced around a straight current-carrying conductor. If the conductor is held in the right hand with the **thumb pointing in the direction of the current**, the **curled fingers indicate the direction of the magnetic field lines** surrounding the conductor. Therefore, this pair is correctly matched.
- **Pair 2 is correct:** **Fleming's Left-hand Rule** is used to determine the direction of the **force (or motion)** experienced by a current-carrying conductor placed in a magnetic field. When the thumb, forefinger, and middle finger of the left hand are held mutually perpendicular, the **forefinger indicates the direction of the magnetic field**, the **middle finger indicates the direction of the current**, and the **thumb gives the direction of the force or motion**. Hence, this pair is correctly matched.
- **Pair 3 is correct:** **Fleming's Right-hand Rule** is applied in the case of **electromagnetic induction** to determine the direction of the **induced current**. If the thumb, forefinger, and middle finger of the right hand are held mutually perpendicular, the **forefinger represents the direction of the magnetic field**, the **thumb indicates the motion of the conductor**, and the **middle finger gives the direction of the induced current**. Therefore, this pair is correctly matched.
- **Pair 4 is correct:** **electromagnetic induction** is the phenomenon in which an **electromotive force (emf)** is induced in a conductor whenever there is a **change in the magnetic field linked with it**. In other words, a changing magnetic flux through a conductor or coil induces an emf, and if the circuit is closed, an induced current flows. Therefore, this pair is also correctly matched.

Q 62. Ans. D

Explanation:

- The International System of Units (SI) consists of seven base units, from which all other physical units are derived. These base units are:

Physical Quantity	SI Base Unit
Length	metre (m)
Mass	kilogram (kg)
Time	second (s)
Electric current	ampere (A)
Thermodynamic temperature	kelvin (K)
Amount of substance	mole (mol)
Luminous intensity	candela (cd)

Q 63. Ans. D

Explanation:

- The effect of a force depends not only on its magnitude but also on the **time duration** for which it acts. The concept of **impulse** is defined as the product of force and the time for which the force acts.
✓ **Impulse = Force × Time**
- For a given change in momentum, if the **time of contact is very short**, the force produced is **very large**. A karate player strikes the tiles with a very fast hand movement, making the **time of impact extremely small**. As a result, a **large force** acts on the tiles, causing them to break.

Q 64. Ans. D

Explanation:

- Static friction** acts between two surfaces that are in contact but **do not move relative to each other**. It prevents the object from moving when an external force is applied. As long as the object remains at rest, static friction adjusts its magnitude to balance the applied force. Therefore, static friction is the **friction without motion**, making the matching **Static Friction** → 1 correct.
- The applied force on a stationary object increases, the static friction also increases until it reaches its **maximum value**. This maximum value of static friction is called **limiting friction**. If the applied force exceeds this limit, the object starts moving. Thus, limiting friction is the **maximum friction acting just before motion begins**, making the matching **Limiting Friction** → 3 correct.
- Once an object starts moving over another surface, the friction acting between the surfaces is called **kinetic (or sliding) friction**. This friction opposes the motion of the object and is generally **less than limiting friction**. Therefore, kinetic friction is the

friction during motion, making the matching **Kinetic Friction** → 4 correct.

- Rolling friction** acts when an object rolls over a surface. It is much smaller than sliding or kinetic friction, which is why wheels and ball bearings are used in vehicles and machines to reduce friction. Since rolling friction is the **least among the common types of friction**, the matching **Rolling Friction** → 2 is correct.

Q 65. Ans. A

Explanation:

- Statement 1 is correct:** Haemophilia is a **genetic disorder** in which the normal process of **blood clotting is impaired** due to the absence or deficiency of specific clotting factors. As a result, even a minor injury can lead to prolonged bleeding because blood does not clot efficiently. haemophilia as one of the well-known **Mendelian disorders** affecting the blood-clotting mechanism. Therefore, the statement is correct.
- Statement 2 is incorrect:** Haemophilia is **not** inherited as an autosomal dominant trait. Instead, it is an **X-linked recessive disorder**. The defective gene responsible for haemophilia is located on the **X chromosome**. Since males possess only one X chromosome, they express the disorder if they inherit the defective gene. Females generally have two X chromosomes and usually become **carriers** unless they inherit the defective gene from both parents, which is rare. Therefore, the statement is incorrect.

Q 66. Ans. D

Explanation:

- Assertion (A) is False:** A catalyst **does not change the equilibrium position** or the equilibrium composition of a reaction. A catalyst only helps the system **reach equilibrium faster**.
✓ A catalyst
 - does **not** alter the equilibrium constant,
 - does **not** shift the equilibrium to either side,
 - only decreases the time required to attain equilibrium.
- ✓ Therefore, the Assertion is **false**.
- Reason (R) is true:** a catalyst provides an alternative reaction pathway with a lower activation energy, thereby increasing the rate of the reaction. It accelerates both the forward and the backward reactions to a similar extent. As a result, the

system reaches equilibrium more quickly, but the equilibrium position and the equilibrium constant remain unchanged because the catalyst does not favour either the reactants or the products.

Q 67. Ans. B

Explanation:

- A **P-type semiconductor** is obtained by doping a pure semiconductor such as **silicon (Si)** or **germanium (Ge)** with **trivalent impurity atoms**. Trivalent elements, such as **boron (B)**, **aluminium (Al)**, **gallium (Ga)**, and **indium (In)**, have **three valence electrons**. When these impurities replace a silicon or germanium atom in the crystal lattice, one covalent bond remains incomplete, creating a **hole**. These holes act as the **majority charge carriers** in a P-type semiconductor, while electrons become the minority carriers. Therefore, a semiconductor doped with trivalent elements is called a **P-type semiconductor**.

Q 68. Ans. B

Explanation:

- **Colour blindness** is an **X-linked recessive genetic disorder**. The gene responsible for colour blindness is located on the **X chromosome**. Since males possess only one X chromosome (XY), they express the disorder if they inherit the defective gene from their mother. Females have two X chromosomes (XX), so they generally become **carriers** if only one X chromosome carries the defective gene and are affected only if both X chromosomes possess the mutant allele. Therefore, the gene responsible for colour blindness is located on the **X chromosome**.

Q 69. Ans. A

Explanation:

- **Sickle cell anemia** is a **Mendelian genetic disorder** caused by a **point mutation** in the gene encoding the θ -chain of haemoglobin. Specifically, the disease results from the **substitution of a single nucleotide** in the DNA sequence of the θ -globin gene. This single base substitution changes the codon from **GAG to GUG** in the messenger RNA, leading to the replacement of **glutamic acid** by **valine** at the sixth position of the θ -globin polypeptide chain. This altered haemoglobin (HbS) causes red blood cells to assume a sickle shape under low oxygen conditions. Therefore, sickle cell anaemia is caused by the **substitution of one nucleotide**.

Q 70. Ans. D

Explanation:

- The **sarcomere** is the **functional unit of muscle contraction**. It is the segment of a myofibril located **between two successive Z-lines (Z-discs)**. A muscle fibre contains numerous myofibrils, and each myofibril is made up of many sarcomeres arranged end to end. During muscle contraction, the **actin (thin) filaments slide over the myosin (thick) filaments**, causing the sarcomere to shorten. This shortening of thousands of sarcomeres simultaneously results in the contraction of the entire muscle. Therefore, the **sarcomere is regarded as the structural and functional unit responsible for muscle contraction**.

Q 71. Ans. C

Explanation:

- **Myasthenia gravis** is an **autoimmune disorder** that affects the **neuromuscular junction**, the site where a motor neuron communicates with a skeletal muscle fibre. In this disorder, antibodies are produced against the **acetylcholine receptors** present on the muscle membrane at the neuromuscular junction. As a result, nerve impulses are not transmitted effectively to the muscles, leading to **muscle weakness and fatigue**, especially after repeated activity. Therefore, Myasthenia gravis primarily affects the **neuromuscular junction**.

Q 72. Ans. C

Explanation:

- The **testes** are located outside the abdominal cavity in a pouch called the **scrotum**. The scrotum maintains the testes at a temperature that is **2–2.5°C lower than the normal internal body temperature**, which is essential for the normal process of **spermatogenesis (sperm production)**. This slightly lower temperature provides the optimum conditions required for the development and maturation of sperm. Therefore, the scrotum maintains the temperature of the testes at approximately **2–3°C below body temperature**.

Q 73. Ans. C

Explanation:

- **Thyroid-Stimulating Hormone (TSH)** is secreted by the **anterior pituitary gland** and regulates the activity of the **thyroid gland**. TSH stimulates the thyroid gland to synthesize and secrete the thyroid

hormones **thyroxine (T₄)** and **triiodothyronine (T₃)**. These hormones play an important role in regulating **basal metabolic rate (BMR)**, growth, development, and metabolism of carbohydrates, proteins, and fats. Therefore, the hormone that stimulates the thyroid gland is TSH.

Q 74. Ans. D

Explanation:

- **Erepsin** is a group of **intestinal peptidases** present in the **intestinal juice (succus entericus)** secreted by the small intestine. Erepsin acts on **peptides and polypeptides** produced during protein digestion and converts them into their final products, **amino acids**. These amino acids are then absorbed through the intestinal villi into the bloodstream. Therefore, erepsin converts **polypeptides into amino acids**.

Q 75. Ans. D

Explanation:

- **Statement 1 is correct:** *Wuchereria bancrofti* is a **filarial worm** that causes **filariasis (elephantiasis)** in humans. The disease affects the **lymphatic vessels and lymph nodes**, leading to chronic inflammation and obstruction of lymph flow. As a result, affected body parts, particularly the lower limbs and genital organs, may become swollen. Therefore, the statement is correct.
- **Statement 2 is correct:** the **adult worms of Wuchereria bancrofti** live in the **lymphatic vessels**, where they interfere with the normal flow of lymph. Their presence causes blockage and inflammation of the lymphatic system, which is responsible for the characteristic symptoms of lymphatic filariasis. Therefore, the statement is correct.
- **Statement 3 is correct:** **female mosquitoes** act as the **intermediate hosts and vectors** of *Wuchereria bancrofti*. When a mosquito feeds on the blood of an infected person, it ingests **microfilariae**, which undergo development inside the mosquito. During a subsequent blood meal, the infective larvae are transmitted to another human host, completing the life cycle. Therefore, the statement is correct.

Q 76. Ans. D

Explanation:

- **Pair 1 is correct:** **epilepsy** is a neurological disorder characterised by **recurrent episodes of abnormal and excessive electrical activity of neurons in the brain**. This abnormal neuronal activation leads to seizures, which may involve loss of consciousness,

involuntary muscle contractions, or abnormal behaviour. Therefore, the pair is correctly matched.

- **Pair 2 is correct:** **Parkinson's disease** is a progressive disorder of the central nervous system caused by the **degeneration of dopamine-secreting neurons** in the brain. The resulting **deficiency of dopamine** leads to symptoms such as resting tremors, muscular rigidity, slow movements (bradykinesia), and impaired posture and balance. Therefore, this pair is correctly matched.
- **Pair 3 is correct:** a **stroke** occurs when the **blood supply to a part of the brain is interrupted** due to either the **rupture of a blood vessel (brain haemorrhage)** or the **blockage of a blood vessel by a clot**. In both cases, brain cells are deprived of oxygen and nutrients, leading to neurological deficits. Therefore, the pair is correctly matched.
- **Pair 4 is correct:** **Huntington's disease** as a **hereditary neurodegenerative disorder** caused by a **genetic mutation**. The disease is inherited in an autosomal dominant manner and results in progressive degeneration of nerve cells, leading to involuntary movements, cognitive decline, and behavioural changes. Therefore, this pair is correctly matched.

Q 77. Ans. B

Explanation:

- The **Moonshot Project** is an advanced research initiative aimed at developing **AI-powered brain co-processors** that can interface with the human brain to decode neural signals, process them using artificial intelligence, and deliver targeted stimulation to restore lost neurological functions.
- It integrates **neuromorphic computing, artificial intelligence, neuroscience, and bioelectronics** to create **closed-loop brain-machine systems** capable of continuously sensing, processing, and responding to neural activity.
- The project is being led by the **Indian Institute of Science (IISc)**.
- It is funded by the **Pratiksha Trust**, established by **Kris Gopalakrishnan and Sudha Gopalakrishnan**.
- Its primary objective is to develop **indigenous AI-driven neurotechnology** that can restore **cognitive and motor functions**, particularly in patients suffering from neurological conditions such as **stroke**.
- The initiative also seeks to create **cost-effective, clinically deployable brain-machine technologies** tailored to India's healthcare needs, with potential applications in other low-resource healthcare systems worldwide.

Q 78. Ans. A

Explanation:

- **Tokens** are the fundamental units of data processed by **Large Language Models (LLMs)**. They act as the basic building blocks of AI language by converting human-readable text into a numerical format that computers can understand. When a user enters a prompt, the text is first broken down into smaller units called **tokens** through a process known as **tokenisation**. Each token is then assigned a unique numerical identifier (token ID), enabling the model to process the input mathematically. Based on patterns learned during training, the LLM predicts the most likely next token in the sequence to generate a coherent response. Finally, the predicted token IDs are converted back into human-readable text through **detokenization**, producing the output seen by the user.

Q 79. Ans. A

Explanation:

- **Statement 1 is correct:** PSLV-C62 was planned to launch EOS-N1 (Anvesha) for the **Defence Research and Development Organisation (DRDO)**. However, during the mission, a **roll-rate disturbance** occurred in the **third stage (PS3)**, causing the rocket to lose control and deviate from its intended trajectory, resulting in mission failure.
- **Statement 2 is incorrect:** The mission carried **15 commercial co-passenger satellites**, not 25. Due to the third-stage anomaly, **all 16 satellites** (EOS-N1 and the 15 commercial payloads) were lost.
- The **Indian Space Research Organisation (ISRO)** has come under scrutiny following two consecutive mission failures of the **Polar Satellite Launch Vehicle (PSLV)**—PSLV-C61 in **May 2025** and PSLV-C62 in **January 2026**. In both missions, the failures were traced to anomalies in the **third stage (PS3)** of the launch vehicle.

Q 80. Ans. C

Explanation:

- **Pair 1 is correct:** Tokenisation is the process of breaking down text into smaller units called tokens (words, sub-words, or characters), which are then converted into numerical form for the model to process. This is the fundamental first step in how language models handle text.
- **Pair 2 is correct:** Fine-tuning is the process of taking a pre-trained model and further training it on a

smaller, task-specific dataset so that it performs better on a particular application (for example, legal text analysis or medical question-answering).

- **Pair 3 is correct:** Hallucination refers to an AI model generating information that is false, fabricated, or not grounded in its training data or the given input, while presenting it as if it were factual. It is a well-known limitation of large language models.
- **Pair 4 is incorrect:** Inference is not the initial training of a model. Inference is the process by which a trained model is used to generate outputs or make predictions on new input (i.e., the model being used after training). The description given (“training a model on a large dataset for the first time”) actually describes pre-training, not inference. So this pairing is wrong.

Q 81. Ans. C

Explanation:

- The Indian Councils Act of 1892 introduced the principle of election in an indirect manner, by providing that some of the non-official members of the councils were to be nominated on the recommendation of certain bodies, although the word election was not used and the system was limited and indirect.
- The Act increased the functions of the legislative councils by empowering them to discuss the annual budget and to address questions to the executive, thereby enlarging the deliberative role of the members, though they could not vote on the budget or move resolutions.
- The Act of 1892 also enlarged the size of the legislative councils, both central and provincial, by increasing the number of additional or non-official members. It represented a modest advance in the association of Indians with the legislative process and prepared the ground for the more substantial reforms of 1909, which introduced separate electorates.

Q 82. Ans. B

Explanation:

- The Constitution makes a synthesis of the British principle of parliamentary sovereignty and the American principle of judicial supremacy, so that the Supreme Court can declare a parliamentary law unconstitutional through judicial review, while Parliament can amend the major portion of the Constitution through its constituent power.
- The Indian Parliament does not enjoy the unlimited

sovereignty of the British Parliament, since it functions within a written constitution, is subject to judicial review, must respect the fundamental rights, and operates within a federal scheme; the American doctrine of judicial supremacy is itself limited in India by the use of the expression procedure established by law.

- This balance reflects the broader character of the Constitution, which borrows the parliamentary form of government from Britain, the fundamental rights and judicial review from the United States, the directive principles from Ireland, and the emergency provisions from Germany and the Government of India Act of 1935, integrating them into a coherent constitutional scheme suited to Indian conditions.

Q 83. Ans. A

Explanation:

- Article 35 provides that the power to make laws to give effect to certain fundamental rights shall vest only in Parliament and not in the state legislatures, for example, laws prescribing residence as a condition for certain employments, laws empowering courts other than the Supreme Court and High Courts to issue writs, and laws relating to the armed forces and martial law under Articles 33 and 34.
- Article 21A, inserted by the 86th Amendment Act of 2002, makes free and compulsory education a fundamental right only for children of the age of six to fourteen years, and not for all citizens up to the age of eighteen; the corresponding fundamental duty under Article 51A obliges a parent or guardian to provide opportunities for education to a child in that age group.
- The Right of Children to Free and Compulsory Education Act of 2009 was enacted to give effect to Article 21A, providing for free and compulsory education in a neighbourhood school and reserving a proportion of seats in private unaided schools for children from disadvantaged groups, thereby translating the directive principle of early education into an enforceable right.

Q 84. Ans. A

Explanation:

- **Statement 1 is correct:** Article 44 directs the State to endeavour to secure for the citizens a uniform civil code throughout the territory of India, an objective intended to replace the diverse personal

laws governing matters such as marriage, divorce, inheritance, and succession with a common set of civil laws applicable to all citizens.

- **Statement 2 is incorrect:** A uniform civil code is a directive principle and is non-justiciable, so it cannot itself be enforced through a court; even if enacted, it would be an ordinary law of the legislature and not a fundamental right, although the courts have on several occasions urged the State to take steps towards its realisation.
- Article 44 belongs to the liberal intellectual group of directive principles. The matter remains contentious because personal laws are linked to religious freedom under Articles 25 to 28, and the framers therefore left the uniform civil code as a goal to be achieved gradually through consensus rather than as an immediate and enforceable obligation.

Q 85. Ans. B

Explanation:

- The President holds office for a term of five years from the date of entering upon the office, is eligible for reelection, and can resign by addressing the resignation to the Vice-President. Under Article 59, the President must not be a member of either House of Parliament or of a state legislature, must not hold any other office of profit, and is entitled to emoluments, allowances, and the use of an official residence without payment of rent.
- Under Article 60, the oath of office to the President is administered by the Chief Justice of India, and in his absence by the senior-most judge of the Supreme Court available, before the President enters upon the office.
- In the oath, the President swears to faithfully execute the office, and to the best of his ability to preserve, protect, and defend the Constitution and the law, and to devote himself to the service and well-being of the people of India, a formulation that underlines his role as the guardian of the Constitution.

Q 86. Ans. A

Explanation:

- **Statement 1 is correct:** Under Article 75(3), the Council of Ministers is collectively responsible to the Lok Sabha, which means that the ministers sink and swim together and that the Lok Sabha can remove the entire ministry by passing a resolution of no confidence, even though the principle of collective responsibility binds the cabinet decisions of all ministers.

- **Statement 2 is correct:** The principle of individual responsibility, embodied in Article 75(2), provides that the ministers hold office during the pleasure of the President, which means that the President can remove a minister, though only on the advice of the Prime Minister, when there is a difference of opinion or dissatisfaction with the performance of the minister.
- **Statement 3 is incorrect:** There is no procedure for a no-confidence motion against a single minister; a no-confidence motion is directed against the entire Council of Ministers, and if it is passed, the whole Council, and not merely one minister, must resign. The Council of Ministers is the real executive authority, while the President is the nominal executive.
- **Statement 2 is incorrect:** In certain situations the Governor cannot promulgate an ordinance without the prior instruction of the President, namely where a bill containing the same provisions would have required the previous sanction of the President for its introduction, where the Governor would have deemed it necessary to reserve a bill containing those provisions for the consideration of the President, and where an act containing those provisions would have been invalid without the President's assent.
- An ordinance issued by the Governor must be laid before the state legislature when it reassembles and ceases to operate at the expiry of six weeks from the reassembly, unless it is approved earlier, and it can also be withdrawn at any time by the Governor. The ordinance-making power is coextensive with the legislative power of the state legislature and is intended to meet situations requiring immediate action when the legislature is not in session.

Q 87. Ans. B

Explanation:

- The Hussainara Khatoon case of 1979, which concerned the plight of a large number of undertrial prisoners languishing in the jails of Bihar, is regarded as the first reported public interest litigation in India, and it led to the release of many undertrials and the recognition of the right to a speedy trial as part of Article 21.
- The S. P. Gupta case of 1981 is closely related to the development of public interest litigation, for in that case the Supreme Court formally articulated the concept and liberalised the rule of locus standi, holding that any member of the public acting bona fide could move the court for the enforcement of the rights of those who could not approach it themselves.
- Public interest litigation was promoted in India largely by Justices P. N. Bhagwati and V. R. Krishna Iyer from the late 1970s, drawing on the American experience. It can be filed under Article 32 before the Supreme Court or under Article 226 before a High Court, and the courts have laid down guidelines to ensure that it is used for genuine public causes and not for private, political, or publicity interests.

Q 88. Ans. A

Explanation:

- **Statement 1 is correct:** Under Article 213 the Governor can promulgate ordinances when the state legislative assembly, or either House in a state having a Legislative Council, is not in session, and when circumstances exist that render it necessary to take immediate action; such an ordinance has the same force and effect as an act of the state legislature.

Q 89. Ans. C

Explanation:

- Article 344 provides for the appointment by the President of an Official Language Commission, at the expiration of five years and again at the expiration of ten years from the commencement of the Constitution, to make recommendations regarding the progressive use of Hindi and the restrictions on the use of English for the official purposes of the Union.
- Article 344 also provides for the constitution of a Committee of Parliament on Official Language, consisting of thirty members, twenty from the Lok Sabha and ten from the Rajya Sabha, to examine the recommendations of the Official Language Commission and to report its opinion to the President.
- The first Official Language Commission was appointed in 1955 under the chairmanship of B. G. Kher. Article 343 declares Hindi in the Devanagari script to be the official language of the Union, with the continued use of English, and the Official Languages Act of 1963 provides for the indefinite continuation of English alongside Hindi for the official purposes of the Union and for use in Parliament.

Q 90. Ans. A

Explanation:

- **Statement 1 is correct:** The Delimitation Commission readjusts the division of each state into territorial constituencies for elections to the

Lok Sabha and the state legislative assemblies, on the basis of the latest census figures, so as to ensure a fair distribution of seats and a roughly equal population per constituency.

- **Statement 2 is incorrect:** The orders of the Delimitation Commission have the force of law and cannot be called in question before any court, a provision intended to prevent elections from being held up indefinitely by litigation over the boundaries of constituencies; the orders are laid before the Lok Sabha and the concerned state assemblies, but cannot be modified by them.
- Under Article 82 the allocation of seats and the division into constituencies are to be readjusted after each census, but the allocation of seats in the Lok Sabha to the states has been frozen on the basis of the 1971 census, and the division of states into constituencies on the basis of the 2001 census, until the first census taken after the year 2026, in order not to penalise states that have controlled their population.

Q 91. Ans. B

Explanation:

- **Infrastructure Finance Companies (IFCs):** These are a category of Non-Banking Financial Companies (NBFCs) that primarily engage in infrastructure financing. To qualify as an IFC, a company must deploy at least 75% of its total assets in infrastructure loans.
- **Core Investment Companies (CICs):** These are NBFCs that carry on the business of acquisition of shares and securities, which hold not less than 90% of its Net Assets in the form of investment in equity shares, preference shares, bonds, debentures, debt, or loans in group companies.
- **Housing Finance Companies (HFCs):** These are companies whose principal business is providing finance for housing. As of 2019, the regulatory oversight for HFCs was transferred from the National Housing Bank (NHB) to the Reserve Bank of India (RBI), and they are currently classified as a specialised category of NBFCs.

- **Payment Banks:** Payment Banks are not NBFCs. They are **Banks** licensed by the Reserve Bank of India under the Banking Regulation Act, 1949. While they have restricted operations (they cannot lend or issue credit cards), they hold a full banking license, which distinguishes them from NBFCs.
- **RBI broadly categorises NBFCs into the following specific buckets based on their operational focus:**
 - ✓ **Investment and Credit Company (NBFC-ICC):** (Formed by merging the old Asset Finance, Loan, and Investment company categories).
 - ✓ **Infrastructure Finance Company (NBFC-IFC)**
 - ✓ **Core Investment Company (NBFC-CIC)**
 - ✓ **Housing Finance Company (NBFC-HFC)**
 - ✓ **Infrastructure Debt Fund (NBFC-IDF)**
 - ✓ **NBFC-Micro Finance Institution (NBFC-MFI)**
 - ✓ **NBFC-Factor** (Engaged in the principal business of factoring/receivables).
 - ✓ **Peer-to-Peer Lending Platform (NBFC-P2P)**
 - ✓ **Account Aggregator (NBFC-AA)**

Q 92. Ans. B

Explanation:

- **Bonds (2) - Lowest Risk:** Bonds are generally debt instruments issued by governments or highly reputable corporations. They are often secured against specific assets. Because bondholders are creditors with a priority claim on assets, they are the first to be paid in the event of default.
- **Debentures (1) - Moderate Risk:** Debentures are also debt instruments, but they are typically **unsecured** (not backed by physical collateral). They rely on the issuer's creditworthiness. While they still rank higher than shareholders, they carry slightly more risk than secured bonds.
- **Ordinary Shares (3) - Highest Risk:** Equity holders are the "residual claimants." If a company goes bankrupt, shareholders are the last to be paid—only after all bonds, debentures, taxes, and other liabilities have been satisfied. They bear the full impact of market volatility and performance fluctuations.

Financial Instrument	Type	Collateral/Backing	Priority during Bankruptcy	Risk Level
Bonds	Debt	Secured by specific assets / Government backing	First Priority	Lowest
Debentures	Debt	Unsecured (backed by company's reputation)	Second Priority (General Creditors)	Medium
Ordinary Shares	Equity	Unsecured (Ownership Capital)	Last Priority (Residual Claimants)	Highest

Q 93. Ans. D**Explanation:**

- To calculate NNP at factor cost from NNP at market prices, indirect taxes are deducted and subsidies are added.
- **Equation:** $\text{NNP at factor cost} = \text{NNP at market prices} - \text{Indirect taxes} + \text{Subsidies}$
 - ✓ Market prices include indirect taxes, which are collected by the government rather than the producers. Subtracting these taxes reveals the actual production cost. Subsidies reduce the market price of goods. Adding subsidies back restores the value to represent the true cost incurred during production.

Q 94. Ans. A**Explanation:**

- In India, the National Sample Survey Office (NSSO), and now the Periodic Labour Force Survey (PLFS), measures employment and unemployment using three distinct reference periods.
- The Usual Status (US) approach is determined with reference to a period of 365 days (one year) preceding the date of the survey.
 - ✓ Under this approach, a person is classified based on the activity they pursued for a relatively long time (major time criterion, generally 183 days or more) during that one-year period. It is further divided into two categories:
- **Principal Status (PS):** The activity on which the person spent the majority of the 365 days.
- **Subsidiary Status (SS):** Any economic activity pursued by a person for a period of 30 days or more during the 365 days, even if their principal status was unemployed or outside the labour force (like a student).

Q 95. Ans. A**Explanation:**

- Hidden or Disguised Unemployment occurs when more people are engaged in a job than are actually required to perform it. Because the workforce is oversaturated for that specific task, the marginal productivity of the extra workers is essentially zero. If you were to remove these redundant workers, the total output or production would not decrease.
- This type of unemployment is overwhelmingly prevalent in the Indian agricultural sector, where an entire joint family might work on a small plot

of land that realistically only needs two or three people to farm it efficiently.

Type of Unemployment	Core Cause / Definition	Example
Structural	Skill Mismatch: A mismatch between the skills the workers possess and the skills demanded by the current economy/ employers.	Coal miners losing jobs because the energy sector shifted to renewable energy.
Cyclical	Economic Downturn: Tied to the boom-and-bust phases of the business cycle. Caused by a general decline in macroeconomic demand.	Mass layoffs during the 2008 Financial Crisis or the COVID-19 pandemic.
Voluntary	Personal Choice: Workers who are physically able to work but choose not to at the prevailing wage rates.	A person quitting a low-paying job because they prefer to live on their savings while looking for a higher salary.

Q 96. Ans. D**Explanation:**

- In economics, the factors of production are the fundamental building blocks of any economy. They are the inputs required for the creation of all goods and services. Classical economics categorizes these resources into four distinct pillars:
 - ✓ **Land:** This includes not just real estate, but all **naturally occurring resources** used in production. Examples include agricultural soil, minerals, water, forests, and oil.
 - ✓ **Labour:** This represents the **human effort** (both physical and mental) contributed to the production process. A construction worker, a software engineer, and a teacher all provide labor.
 - ✓ **Capital:** This refers to **man-made physical assets** used to produce other goods and services. It includes machinery, tools, factory buildings, delivery vehicles, and computers. (Note: In this context, it means physical capital, not financial capital/money, since money itself doesn't produce anything).

- ✓ **Entrepreneurship:** This is the human element that **combines the other three factors**. An entrepreneur takes on the financial risk, makes the operational decisions, and innovates to bring a product or service to the market.

Q 97. Ans. B

Explanation:

- Statement 1 describes a measure that calculates the weighted average value of a currency against a basket of currencies of its major trading partners, using trade weights to reflect the relative importance of each partner. This is the defining characteristic of an effective exchange rate measure, as opposed to a bilateral exchange rate which only compares two currencies.
- Statement 2 specifies that this measure is nominal, meaning it does not adjust for inflation differentials or price level differences between the home country and its trading partners. It captures only the movement in currency values as observed in foreign exchange markets.
- Together, both statements point to the Nominal Effective Exchange Rate (NEER), which is the trade-weighted average of a currency's nominal exchange rates against a basket of currencies, without any inflation adjustment.
- REER (Real Effective Exchange Rate) adjusts NEER for relative price level differences between countries, making it a better indicator of international competitiveness. PPP Exchange Rate is based on the idea that identical goods should cost the same across countries when expressed in a common currency. Nominal Exchange Rate refers simply to the bilateral exchange rate between two currencies at a given point in time, without any trade-weighting or basket comparison.

Q 98. Ans. C

Explanation:

- **Horizontal FDI (Market-Seeking)**
 - ✓ **Description:** The firm opens a subsidiary in a foreign country to perform the same activities it does in its home country.
 - ✓ **Why:** The goal is usually to bypass trade barriers (like tariffs), reduce transportation costs, or get closer to local customers.
 - ✓ **Example:** A U.S.-based automobile manufacturer building an assembly plant in India to produce the same car models it sells in the U.S.

● Vertical FDI (Efficiency/Resource-Seeking)

- ✓ **Description:** The firm invests in a business that is at a different stage of the same production chain.
 - **Backward Vertical FDI:** Investing in suppliers (e.g., an electronics company buying a foreign copper mine to secure raw materials).
 - **Forward Vertical FDI:** Investing in distributors or retailers (e.g., a clothing manufacturer acquiring foreign retail stores to sell their products directly).
- ✓ **Why:** The goal is to control the supply chain, lower costs, or ensure supply security.

● Conglomerate FDI (Diversification-Seeking)

- ✓ **Description:** A company invests in a foreign business that is unrelated to its current business activities.
- ✓ **Why:** The primary goal is diversification. By moving capital into a completely different industry or market, the company reduces its risk; if its home industry faces a downturn, the foreign business might remain profitable.
- ✓ **Example:** A major telecommunications company investing in a foreign hospitality or real estate chain.

Q 99. Ans. C

Explanation:

- The Sundarbans Delta, also widely known as the Ganges-Brahmaputra Delta, is the largest river delta in the world. It is formed by the confluence of the Ganges, Brahmaputra, and Meghna rivers as they flow into the Bay of Bengal. This vast region spreads across both India and Bangladesh. It is celebrated globally not only for its size but also for the Sundarbans mangrove forest, which is the largest of its kind on the planet.
- **Additional Facts:**
 - ✓ **Geographic Significance:** This delta is a low-lying region prone to tidal action. The network of distributaries creates a complex ecosystem that is vital for biodiversity.
 - ✓ **Ecological Importance:** The mangrove forests act as a natural barrier against storms and cyclones, protecting the inland areas. This is why the region is considered environmentally sensitive and protected as a UNESCO World Heritage site.
 - ✓ **Comparison to other deltas:**
 - ✓ **Nile Delta (Egypt):** Known for being an arcuate

(fan-shaped) delta, but significantly smaller in area than the Ganges-Brahmaputra system.

- ✓ **Mekong Delta (Vietnam):** A highly productive agricultural region but smaller in geographic scale compared to the Sundarbans.
- ✓ **Mississippi Delta (USA):** Famous for its bird-foot shape, which is structurally distinct from the depositional patterns of the Sundarbans.

Q 100. Ans. B

Explanation:

- **Atacama Desert:** This is located in **South America** (primarily in Chile), not Africa. It is famous for being one of the driest non-polar places on Earth.
- **Great Victoria Desert:** This is located in **Australia**. It is the largest desert in Australia.
- **Sonoran Desert:** This is located in **North America** (straddling the border of the United States and Mexico).

Q 101. Ans. B

Explanation:

- The Ionosphere is a series of regions in parts of the mesosphere and thermosphere where high-energy solar radiation strips electrons from atoms and molecules, creating a plasma of charged particles (ions).
- When radio waves are transmitted from the ground, they travel upward into the Ionosphere. The dense concentration of free electrons causes the radio waves to refract and “bend” back toward the Earth’s surface.
- Because the signal is reflected off this layer, it can travel far beyond the horizon, curving around the Earth’s surface. This process is known as Skywave propagation or “radio skipping,” which allows for communication across thousands of kilometers without the need for line-of-sight cables or satellites.

Q 102. Ans. C

Explanation:

- **Statement 1 is correct:** The Mahi River originates in the Vindhya Range in Madhya Pradesh. It flows in a northward direction, enters Rajasthan, and then turns south towards Gujarat. Because of this curved path, it crosses the Tropic of Cancer twice. It is the only major river in India to exhibit this specific pattern.
- **Statement 2 is correct:** Udaipur, located in the Gomati district of Tripura, is situated extremely close to the Tropic of Cancer. In the context of

Indian competitive examinations, it is the standard answer for the city located nearest to the Tropic of Cancer.

● Additional Facts:

- ✓ **The Eight States:** The Tropic of Cancer passes through exactly eight Indian states. Memorizing these is essential: Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura, and Mizoram.
- ✓ Examiners often ask which state capital is closest to the Tropic of Cancer. Gandhinagar (Gujarat) and Ranchi (Jharkhand) are often cited as being very close to the line. Be careful not to confuse a “city” like Udaipur with a “state capital.”
- ✓ The Tropic of Cancer represents the northernmost latitude where the sun can appear directly overhead at noon. In India, this line serves as a divider between the tropical climate zone to the south and the subtropical/temperate climate zone to the north.

Q 103. Ans. D

Explanation:

- A **tombolo** is a narrow strip of land, such as a beach or bar, that connects an island to the mainland or to another island. The formation happens due to wave refraction. As waves approach an island, the island blocks the direct path of the waves. Sediments, such as sand or shingle, accumulate in the protected area behind the island where wave energy is lower. Over time, these sediments build up into a ridge, eventually linking the island to the shore.
- **Lagoon:** A shallow body of water separated from a larger sea by a narrow landform like a sandbar, but it does not connect an island to the mainland in the same way.
- **Estuary:** A partially enclosed coastal body of water where freshwater from rivers meets and mixes with saltwater from the ocean.
- **Fjord:** A long, narrow inlet with steep sides, created by glacial erosion rather than by wave deposition.

Q 104. Ans. A

Explanation:

- **Chilika:** Located in Odisha, this is a brackish water lagoon and the largest lagoon in India.
- **Pulicat:** Located on the border of Andhra Pradesh and Tamil Nadu, this is the second-largest brackish water lagoon in India.
- **Sambhar:** Located in Rajasthan, this is an inland

saline lake. It is not a lagoon because it is not connected to the sea.

- **Wular:** Located in Jammu and Kashmir, this is a freshwater lake primarily formed by tectonic activity, not by coastal wave deposition.

Q 105. Ans. A

Explanation:

- **Laccolith:** These form when magma pushes the overlying strata upward, creating a large, dome-shaped mound.
- **Lopolith:** These form when magma settles into a basin-like structure, resulting in a saucer shape.
- **Phacolith:** These are lens-shaped masses found specifically in the crests of anticlines or the troughs of synclines due to folding.

Igneous Intrusion	Shape
Laccolith	Dome-shaped (Mushroom-shaped)
Lopolith	Saucer-shaped (Basin-like)
Phacolith	Lens-shaped (Wedge-shaped)

Q 106. Ans. B

Explanation:

- A Deranged Drainage Pattern is characterised by a complete lack of organisation in the stream network. It happens when geological forces, specifically glaciation, disrupt the pre-existing landscape.
- Glaciers scrape and reshape the land, leaving behind an irregular surface full of mounds and hollows. When the ice melts, these hollows fill with water, forming lakes and marshes. Because the surface is so uneven and the original river channels have been blocked or filled with debris, the resulting streams flow in many different directions without a clear, connected path.
- Unlike other patterns (like Dendritic or Trellis) which reflect the underlying rock structure, a deranged pattern reflects a “chaotic” history of landscape modification, typical of the Canadian Shield or parts of Scandinavia where glaciers recently retreated.

Q 107. Ans. B

Explanation:

- The primary geological driver for the widening of the Atlantic Ocean is **Seafloor Spreading** occurring at the **Mid-Atlantic Ridge**.
- **Divergent Boundary:** The Mid-Atlantic Ridge is a massive underwater mountain range that runs the length of the Atlantic Ocean. It represents

a **divergent plate boundary**, where the North American and Eurasian plates (in the north) and the South American and African plates (in the south) are moving away from each other.

- **Magma Upwelling:** As these plates move apart, the pressure on the underlying mantle decreases. This allows molten magma to rise from the mantle, fill the gap, and solidify to create new oceanic crust.
- **The “Conveyor Belt”:** This continuous process acts like a conveyor belt, adding new material to the edges of the tectonic plates and effectively pushing the continents further apart, causing the Atlantic Ocean to widen by a few centimeters every year.

Q 108. Ans. B

Explanation:

- **Rainfall:** This is the most significant form of liquid precipitation. It serves as the primary input for the Earth’s hydrological cycle, replenishing rivers, lakes, and groundwater systems, making it the fundamental source of fresh water.
- **Snowfall:** Unlike rain, snow forms when water vapor undergoes deposition (turning directly into ice) in clouds. As these ice crystals fall, they often stick together to form flakes and accumulate in layers, which is crucial for glaciers and long-term water storage in high altitudes.
- **Hail:** These are solid chunks of ice formed in intense thunderstorms (cumulonimbus clouds). Strong updrafts carry water droplets high into freezing altitudes, where they freeze and grow larger before falling as heavy ice pellets.
- **Dew:** This is technically not “precipitation” in the sense that it does not fall from the sky. It is **condensation**. When the temperature of the air drops to the dew point during the night, water vapor in the air condenses onto cool surfaces like leaves and grass.

Q 109. Ans. A

Explanation:

- **Vamsadhara:** This river originates in the Eastern Ghats of Odisha and flows **eastward** to discharge into the Bay of Bengal.
- **Sharavati:** This river originates in the Western Ghats of Karnataka and flows **westward** to discharge into the Arabian Sea.
- **Vaigai:** This river originates in the Western Ghats of Tamil Nadu and flows **eastward** (specifically southeast) to discharge into the Palk Strait, which connects to the Bay of Bengal.

- **Penner:** This river originates in Karnataka and flows eastward through Andhra Pradesh to discharge into the Bay of Bengal.

Q 110. Ans. B

Explanation:

- **Unnat Bihar, Ujjwal Bihar** was the official theme of **Bihar Diwas 2026** (22–24 March 2026). The theme emphasized the vision of a **progressive, prosperous, and brighter Bihar**, highlighting development in education, infrastructure, culture, and public welfare. It was adopted by the Government of Bihar for the state-level celebrations.
- **Bihar Diwas** is celebrated every year on **22 March** to commemorate the creation of the Province of Bihar and Orissa on **22 March 1912**, when it was separated from the Bengal Presidency.

Q 111. Ans. C

Explanation:

- The total geographical area of Bihar is 94,163 square kilometres, as officially reported by the Directorate of Economics & Statistics, Government of Bihar and other Government of Bihar publications. Bihar occupies about **2.86% of India's total geographical area**. **Jharkhand** was carved out of Bihar on 15 November 2000

Q 112. Ans. C

Explanation:

- According to the Water Resources Department, Government of Bihar, **28 out of 38 districts** in Bihar are classified as **flood-prone**. Approximately **68.80 lakh hectares** out of the state's geographical area are vulnerable to floods, making Bihar one of India's most flood-affected states.

Q 113. Ans. C

Explanation:

- According to the Second Bihar State Irrigation Commission (1994), the rivers of Bihar have been classified into 14 river basins to facilitate integrated planning and management of the state's water resources. These 14 river basins are: Ghaghra, Gandak, Burhi Gandak, Bagmati-Adhwara, Kamla-Balan, Kosi, Mahananda, Main Ganga Stem, Karmnasa, Sone, Punpun, Kiul-Harohar, Badua, and Chandan. This basin-wise classification is used by the Water Resources Department, Government of Bihar for irrigation planning, flood management, and sustainable utilization of water resources.

Q 114. Ans. B

Explanation:

- **Statement 1 is incorrect:** The **Ikorla (Ekorla) Weir** is **not** constructed on the **Kiul River**. According to the Bihar River Atlas and records of the Water Resources Department, Government of Bihar, **Ikorla Weir** is constructed on the **Chandan River** in Banka district.
- **Statement 2 is correct:** The **Ikorla Weir** is located in **Banka district** of Bihar. It is one of the irrigation structures maintained under the Water Resources Department.
- **Statement 3 is correct:** It has been built primarily for **irrigation and flood control**. The primary purpose of the **Ikorla Weir** is to support **irrigation**, while weirs under the Water Resources Department also assist in **regulating river flow**, which contributes to local flood management.

Q 115. Ans. A

Explanation:

- **Statement 1 is correct:** **Sama-Chakeva** is a renowned folk festival of the **Mithila region** of Bihar. It is celebrated after **Chhath Puja**, especially in the districts of Mithila, to welcome the arrival of migratory birds. The festival reflects the rich cultural heritage of Mithila through folk songs, rituals, and traditional clay idols.
- **Statement 2 is incorrect:** **Sama-Chakeva** is **not** associated with the **husband-wife relationship**. It symbolizes the **sacred bond between brothers and sisters**. The festival is based on the legend of **Sama (Shyama)**, the daughter of Lord Krishna, and her brother **Chakeva**, highlighting the values of sibling affection, protection, and reunion.

Q 116. Ans. D

Explanation:

- Among the given districts, **Kishanganj, Purnia, and Katihar** share their eastern boundaries with the state of **West Bengal**. Kishanganj is the easternmost district of Bihar and shares its boundary with West Bengal to the east and Nepal to the north, while Purnia and Katihar border West Bengal. On the other hand, **Araria** does **not** share a boundary with West Bengal; it shares an international boundary with **Nepal**. Similarly, **Supaul** borders **Nepal** to the north and the districts of Araria, Saharsa, Madhepura, and Madhubani within Bihar, but it does **not** border West Bengal.

Q 117. Ans. B

Explanation:

- According to the BEAMS Portal of the Water Resources Department, Government of Bihar (based on the Second Bihar State Irrigation Commission, 1994), the important tributaries of the Chandan River are Orhni, Kuldar, and Chatri. Therefore, Orhni (1), Kuldar (2), and Chatri (4) are correct. Kiul (3) is **not** a tributary of the Chandan River; rather, it is an independent river that forms the Kiul–Harohar Basin. Similarly, Trijugi (5) is **not** listed as a tributary of the Chandan River in the official records of the Water Resources Department.

Q 118. Ans. B

Explanation:

- **Assertion (A) is true:** In Bihar, the loo is a hot, dry wind that prevails during the **pre-monsoon season**, mainly in **April and May**. These winds are most intense over the **southwestern and central parts of the state** and are associated with very high daytime temperatures before the onset of the southwest monsoon.
- **Reason (R) is true:** The loo generally blows at a good speed. Owing to its **high temperature and very low humidity**, it causes severe heat stress, dehydration, and discomfort, often leading to heat-related illnesses.
- However, the **Reason does not explain the Assertion**. The assertion concerns the **period during which the loo blows (April–May)**, whereas the reason merely describes its **speed and impact on human comfort**. The occurrence of the loo during April and May is due to intense land heating and pre-monsoon atmospheric conditions, not because of its velocity or its effects on people. Therefore, both statements are true, but the reason is **not** the correct explanation of the assertion.

Q 119. Ans. C

Explanation:

- Bihar Vibhuti, meaning the “Jewel of Bihar,” is a title of honour conferred upon **Dr. Anugrah Narayan Sinha (1887–1957)** in recognition of his outstanding contributions to India’s freedom struggle and the development of Bihar. A distinguished nationalist leader and close associate of **Mahatma Gandhi**, he is widely regarded as one of the principal architects of modern Bihar. Following India’s independence, he served as the

first Deputy Chief Minister and Finance Minister of Bihar, holding these positions from **1946 until his death in 1957**. Working alongside the state’s first Chief Minister, **Sri Krishna Singh**, he played a pivotal role in laying the foundations of Bihar’s administration and development. Their effective governance earned Bihar high praise from the noted American economist **Paul H. Appleby**, who described it as the “**best governed state in India**.”

Q 120. Ans. C

Explanation:

- **Statement 1 is correct:** Yogendra Shukla was a pioneering Indian revolutionary who played a **foundational role in the Hindustan Socialist Republican Association (HSRA)**. He was actively involved in the organization’s efforts to challenge British rule through revolutionary activities and also trained prominent revolutionaries such as **Bhagat Singh and Batukeshwar Dutt**.
- **Statement 2 is incorrect:** Yogendra Shukla was **from Bihar**, not **Uttar Pradesh**. He is remembered as one of Bihar’s most prominent freedom fighters and revolutionary leaders.
- **Statement 3 is correct:** During the **Quit India Movement (1942)**, Yogendra Shukla made a daring **escape from Hazaribagh Central Jail**. After escaping, he carried the ailing **Jayaprakash Narayan** on his shoulders for nearly **124 kilometres** to safety in **Gaya**, successfully evading British forces.

Q 121. Ans. A

Explanation:

- The **Dr. Sachidanand Sinha Prize** is a prestigious honour instituted by the **State Academy of Bihar** to recognize outstanding contributions to **Legal Literature (Law Literature)**. Named after **Dr. Sachchidananda Sinha**—a distinguished lawyer, parliamentarian, and the **first temporary President of the Constituent Assembly of India**—the award commemorates his significant contributions to India’s legal and constitutional development. It is presented to individuals for exceptional published works, scholarly research, or other notable writings that advance the understanding and development of law and legal systems.

Q 122. Ans. B

Explanation:

- **Madhubani painting** is a traditional folk art of the **Mithila region of Bihar and Nepal**, renowned for its intricate geometric and floral motifs, vibrant natural

colours, and depictions of mythology, nature, and social life. Traditionally, women painted **aripana** (floor designs) and **kohabar** (wall paintings) on the mud walls and floors of their homes during festivals, marriages, births, and other auspicious occasions. According to legend, the art originated when **King Janaka** commissioned artists to decorate the city for the wedding of **Sita and Lord Rama**. Over time, Madhubani painting has evolved from portraying religious themes to addressing contemporary social issues, including women's empowerment, equality, and environmental conservation.

- **Sita Devi** was a pioneering Madhubani artist who popularized the **Bharni** style beyond the Mithila region and introduced the art to national and international audiences. In recognition of her contribution, she was awarded the **Padma Shri** in 1981.
- **Godawari Dutta**, who inherited the tradition from her mother, played a significant role in preserving and promoting Madhubani painting. Over a career spanning more than three decades, she trained over **50,000 students and artists** and was honoured with the **Padma Shri** at the age of 93.
- **Mahasundari Devi** emerged as one of the leading exponents of Madhubani art despite limited formal education. She inspired generations of women artists and established the **Mithila Hastashilp Kalakar Audyogiki Sahyog Samiti**, a cooperative society dedicated to supporting Madhubani artists and handicrafts.
- **Malvika Raj**, a contemporary Dalit artist from **Samastipur**, has given Madhubani painting a modern perspective by depicting the life of **Lord Buddha**, **Dr. B. R. Ambedkar**, and other social reformers while retaining the traditional style. Her innovative works have been exhibited internationally, including at the **University of Edinburgh**.

Q 123. Ans. B

Explanation:

- **Statement 1 is incorrect:** Under the **Mukhyamantri Yuva Udyami Yojana**, eligible beneficiaries are provided **financial assistance of up to ₹10 lakh, not ₹50 lakh**, for establishing and operating business ventures. Of this amount, **up to ₹5 lakh (50%)** is provided as a **non-repayable grant**, while the remaining **up to ₹5 lakh** is provided as a loan.
- **Statement 2 is correct:** The **Mukhyamantri Yuva Udyami Yojana** is a component of the **Mukhyamantri Udyami Yojana**, which is implemented by the **Department of Industries**,

Government of Bihar. The scheme aims to promote entrepreneurship and self-employment among eligible individuals in the state.

- **Statement 3 is correct:** The remaining **50% of the financial assistance (up to ₹5 lakh)** is provided as an **interest-free loan**, with a **nominal interest rate of 1% applicable under specific sub-schemes**. The scheme also includes **mandatory entrepreneurial skill development training** to strengthen the capabilities of beneficiaries.

Q 124. Ans. B

Explanation:

- **Statement 1 is correct:** The **Bihar Budget 2026–27** was prepared in line with the **Saat Nischay-3** vision of the Government of Bihar. The budget emphasizes inclusive growth, human development, and sustainable economic progress through targeted investments in key sectors.
- **Statement 2 is incorrect:** The statement is incorrect because **employment, education, women empowerment, agriculture, and infrastructure were among the major focus areas of the Bihar Budget 2026–27**. The budget prioritizes job creation, quality education, women's empowerment, agricultural development, and infrastructure expansion to accelerate the state's socio-economic development.

Q 125. Ans. C

Explanation:

- The **Bihar Budget 2026–27** projects the state's **Gross State Domestic Product (GSDP)** to grow by **15% at current prices** over the revised estimates for 2025–26. According to the budget estimates, Bihar's GSDP is expected to reach **₹13.1 lakh crore** in 2026–27, reflecting the state's focus on accelerating economic growth. For comparison, the **Union Budget** assumes a nominal GDP growth of **10%** for the country in 2026–27.

Q 126. Ans. C

Explanation:

- According to Bihar budget for 2026–27, the **total budget outlay of the Bihar Budget 2026–27** is **₹3,47,590 crore** (Budget Estimate).
- It is important to distinguish between the two expenditure figures reported by PRS:
 - ✓ **Total Budget Outlay (Total Expenditure):** ₹3,47,590 crore
 - ✓ **Net Expenditure (excluding debt repayment):** ₹3,24,925 crore

Q 127. Ans. A

Explanation:

- According to the Census of India 2011, Bihar accounted for approximately 8.6% of India's total population.
 - ✓ Population of Bihar (2011): 10,40,99,452 (about 10.41 crore)
 - ✓ Population of India (2011): 121.09 crore
 - ✓ Bihar's Share in India's Population: Approximately 8.6%
- With over 10.4 crore people, Bihar was the **third most populous state in India**, after Uttar Pradesh and Maharashtra. Despite occupying only about 2.86% of India's geographical area, the state is home to nearly 8.6% of the country's population, making it one of the most densely populated states in India.

Q 128. Ans. C

Explanation:

- According to the Census of India 2011, Bihar recorded a decadal population growth rate of 25.42% during 2001–2011, making it one of the fastest-growing states in the country during the decade.
 - ✓ Population (2001 Census): Approximately 8.30 crore
 - ✓ Population (2011 Census): Approximately 10.41 crore
 - ✓ Decadal Growth Rate (2001–2011): 25.42%
- The high population growth was attributed to factors such as a **high fertility rate, declining mortality, and improvements in healthcare services**. Although Bihar's population continued to grow rapidly, the growth rate declined compared to the previous decade (1991–2001), indicating the beginning of a demographic transition.

Q 129. Ans. C

Explanation:

- According to the Census of India 2011, 88.71% of Bihar's population lives in **rural areas**, while only 11.29% resides in **urban areas**. This makes Bihar one of the **least urbanized states in India**.
 - ✓ Rural Population: 88.71%
 - ✓ Urban Population: 11.29%
 - ✓ Total Population (2011 Census): 10.41 crore (104.1 million)
- The high proportion of rural population reflects Bihar's predominantly agrarian economy, where agriculture and allied activities remain the primary source of livelihood. The low level of urbanization also highlights the need for greater investment in

urban infrastructure, industrial development, and employment opportunities.

Q 130. Ans. A

Explanation:

- Assertion (A) is **true**: The vast plains of Bihar are covered by **deep alluvial deposits** that have been laid down by major river systems such as the **Ganga, Kosi, Gandak, Burhi Gandak, Bagmati, Kamla-Balan, Mahananda, and Son**. These deposits, formed through the transportation and deposition of sediments by rivers, constitute a thick **alluvial mantle of drift origin**, making the plains highly fertile.
- Reason (R) is **true**: The rivers flowing across Bihar frequently deposit **fresh layers of silt, clay, and sand**, especially during floods. This continuous deposition **renews the soil profile**, enriches soil fertility, and is the primary reason why the plains are covered with thick alluvial deposits.
- Since the repeated deposition of sediments by rivers **creates and maintains the alluvial mantle** of Bihar's plains, the **Reason correctly explains the Assertion**.

Q 131. Ans. D

Explanation:

- **CMS-03 — LVM3-M5**
 - ✓ CMS-03 was launched aboard LVM3-M5, ISRO's heavy-lift launch vehicle. The mission is intended to strengthen India's communication satellite capabilities and demonstrates the operational reliability of the LVM3 platform.
- **NISAR — GSLV-F16**
 - ✓ NISAR (NASA-ISRO Synthetic Aperture Radar) was launched using GSLV-F16. It is the world's first Earth observation satellite carrying dual-frequency (L-band and S-band) Synthetic Aperture Radar (SAR), enabling continuous monitoring of Earth's land and ice surfaces.
- **EOS-9 — PSLV-C61**
 - ✓ EOS-9 was launched aboard PSLV-C61. Although the mission was unsuccessful, it was designed as an Earth Observation Satellite to enhance capabilities in agriculture, forestry, disaster management, and environmental monitoring.
- **NVS-02 — GSLV-F15**
 - ✓ NVS-02 was launched using GSLV-F15 as part of the second-generation NavIC (Navigation with Indian Constellation) programme. The NVS series is designed to improve navigation accuracy, continuity of services, and includes indigenous atomic clocks.

Q 132. Ans. C

Explanation:

- **Pair 1 is correctly matched:** According to the Ministry of Electronics and Information Technology (MeitY) and IndiaAI, the official theme chosen for the India AI Impact Summit 2026 is “Sarvajana Hitaya, Sarvajana Sukhaya”, which translates directly to “Welfare for All, Happiness for All”.
- **Pair 2 is correctly matched:** As per the Ministry of Education, the central Theme Pavilion of the New Delhi World Book Fair (NDWBF) 2026 was “Indian Military History: Valour and Wisdom @ 75”, celebrating 75 years of the nation-building role of the Indian Armed Forces since independence.
- **Pair 3 is correctly matched:** According to the Ministry of External Affairs (MEA), the 11th edition of the Raisina Dialogue 2026 was anchored under the civilizational theme “Saṃskāra – Assertion, Accommodation, Advancement”.

Q 133. Ans. B

Explanation:

- The world’s first Smart Island Monitoring System has been launched in Ecuador. The initiative is designed to strengthen real-time monitoring and conservation of island ecosystems by integrating advanced technologies such as Artificial Intelligence (AI), satellite remote sensing, Internet of Things (IoT) sensors, and geospatial analytics.
- The system enables authorities to monitor biodiversity, ecosystem health, climate change impacts, invasive species, pollution, and natural hazards, thereby supporting evidence-based environmental management and sustainable development of island ecosystems.

Q 134. Ans. D

Explanation:

- **Statement 1 is correct:** Venezuela has become India’s third-largest supplier of crude oil, reflecting India’s continued diversification of its energy import basket and increasing purchases from Latin American producers.
- **Statement 2 is correct:** India remains the world’s largest recipient of remittances, driven by its large overseas diaspora. Remittance inflows play a significant role in supporting household incomes, foreign exchange reserves, and the current account.
- **Statement 3 is correct:** India has climbed to the sixth position in global GDP rankings, reflecting

its expanding economic size and sustained growth among major economies.

Q 135. Ans. C

Explanation:

- **CM Mahila Udyami Yojana**
 - ✓ This is a flagship scheme of the Government of Bihar aimed at promoting entrepreneurship among women by providing interest-free financial assistance to establish new enterprises. It is implemented under the Bihar Startup and Entrepreneurship framework.
 - ✓ Mukhyamantri Udyami Yojana in Bihar has separate components such as the CM Mahila Udyami Yojana, CM Yuva Udyami Yojana, and CM SC/ST Udyami Yojana to promote inclusive entrepreneurship.
- **Jal Jeevan Hariyali Abhiyan**
 - ✓ Launched by the Government of Bihar, this campaign focuses on water conservation, afforestation, rainwater harvesting, rejuvenation of traditional water bodies, and climate resilience. It is one of Bihar’s major initiatives to address environmental degradation and climate change.
- **CM Vridhjan Pension Yojana**
 - ✓ This is a Bihar Government social security scheme that provides monthly pension to senior citizens of the state, irrespective of whether they fall below the poverty line.
- **Bal Shramik Vidya Yojana**
 - ✓ This is not a Bihar scheme. It is an initiative of the Government of Uttar Pradesh aimed at rehabilitating child labourers by providing financial assistance to encourage their education.

Q 136. Ans. B

Explanation:

- **World Press Freedom Index 2026 — 157th**
 - ✓ India was ranked 157th in the World Press Freedom Index 2026, published annually by Reporters Without Borders (RSF). The index assesses the level of press freedom in 180 countries and territories.
- **Global Terrorism Index 2026 — 13th**
 - ✓ India was placed 13th in the Global Terrorism Index (GTI) 2026, released by the Institute for Economics & Peace (IEP). The index measures the impact of terrorism based on incidents, fatalities, injuries, and hostage situations.
- **World Happiness Report 2026 — 116th**

- ✓ India ranked 116th in the World Happiness Report 2026, published by the Wellbeing Research Centre at the University of Oxford in partnership with the United Nations Sustainable Development Solutions Network (UN SDSN).

● **Global Firepower Index 2026 — 4th**

- ✓ India secured the 4th position in the Global Firepower Index 2026, reflecting its overall military strength based on factors such as manpower, equipment, logistics, natural resources, and financial capacity.

Q 137. Ans. B

Explanation:

- Goa became the first State in India to establish a 100% Cyber Fraud Response System, ensuring that every reported cyber fraud complaint receives an immediate and coordinated response. The initiative aims to improve the recovery of defrauded funds, enhance inter-agency coordination, and strengthen public confidence in cybercrime redressal mechanisms.
- The system integrates law enforcement agencies, banks, financial institutions, and cybercrime units to facilitate real-time reporting, prompt freezing of fraudulent transactions, and swift investigation.

Q 138. Ans. D

Explanation:

- HDFC Bank emerged as the most valuable brand in India in 2025, reflecting its strong financial performance, extensive customer base, digital banking leadership, and sustained brand value. HDFC Bank is India's largest private sector bank by market capitalisation.
- It was formed through the merger of Housing Development Finance Corporation (HDFC Ltd.) with HDFC Bank, creating one of the world's largest financial services institutions.
- The ranking is based on factors such as brand valuation, financial performance, customer perception, market influence, and future growth potential.

Q 139. Ans. A

Explanation:

- **Statement 1 is correct:** India's first Cheetah Safari has been launched at Kuno National Park in Madhya Pradesh. The safari has been developed to promote eco-tourism and enhance public awareness about cheetah conservation following the reintroduction of cheetahs under Project Cheetah.

- **Statement 2 is incorrect:** The first phase of the All India Tiger Estimation (AITE) 2026 commenced from the Thanthai Periyar Wildlife Sanctuary in Tamil Nadu, not from the Palamu Tiger Reserve.

Q 140. Ans. D

Explanation:

● **Medicine — Shimon Sakaguchi**

- ✓ Shimon Sakaguchi was awarded the 2025 Nobel Prize in Physiology or Medicine for his pioneering discoveries on regulatory T (Treg) cells, which transformed the understanding of immune regulation and autoimmune diseases.

● **Physics — John M. Martinis**

- ✓ John M. Martinis received the 2025 Nobel Prize in Physics for his groundbreaking contributions to the development of superconducting quantum computing.

● **Chemistry — Omar M. Yaghi**

- ✓ Omar M. Yaghi was awarded the 2025 Nobel Prize in Chemistry for his pioneering work on reticular chemistry, including the development of Metal–Organic Frameworks (MOFs) and Covalent Organic Frameworks (COFs).

● **Peace — María Corina Machado**

- ✓ María Corina Machado, a Venezuelan opposition leader, received the 2025 Nobel Peace Prize for her efforts to promote democracy, human rights, and peaceful political change in Venezuela.

Q 141. Ans. A

Explanation:

Given equation:

$$23 \div 196 - 7 \times 4 + 12$$

After changing the sign as per instruction,

$$23 + 196 \div 7 - 4 \times 12$$

$$= 23 + 28 - 48$$

$$= 51 - 48$$

$$= 3$$

Q 142. Ans. C

Explanation:

The logic followed here is, the product of sum of 1st and 2nd digit and 2nd and 3rd digit of the given number gives the resultant number as shown below.

In, 263: 72,

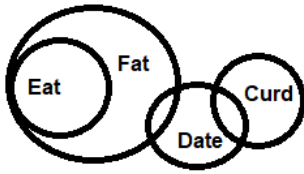
$$(2 + 6) \times (6 + 3) = 8 \times 9 = 72$$

Similarly, in, 258: ?,

$$(2 + 5) \times (5 + 8) = 7 \times 13 = 91$$

Q 143. Ans. D**Explanation:**

Following figure can be formed:



From above figure, none of the conclusions follows
Conclusions:

- I. No Curd are Eat: Doesn't follow (As no direct relation between Curd and Eat, so, no definite conclusion is true)
- II. All Date are Eat: Doesn't follow (As no direct relation between Date and Eat, so, no definite conclusion is true)
- III. Some Fat are not Curd: Doesn't follow (As no direct relation between Fat and curd, so, no definite conclusion is true)

Q 144. Ans. B**Explanation:**

Formula to find the angle between hour hand and minute hand is:

$$\text{Angle} = |(11 \times \text{minutes} - 60 \times \text{hours})/2|^\circ$$

Here, minutes = 54, Hours = 6.

$$\text{Angle} = |(11 \times 54 - 60 \times 6)/2|^\circ$$

$$\text{Angle} = |(594 - 360)/2|^\circ$$

$$\text{Angle} = 234/2 = 117^\circ$$

Q 145. Ans. C**Explanation:**

Let the age of Kush, Sahil and Vikram are $4x$, $7x$ and $5x$ respectively

Sum of Kush and Vikram = 63

$$4x + 5x = 63$$

$$9x = 63$$

$$x = 63 \div 9$$

$$x = 7$$

Present age of Sahil is $7x$

$$7 \times 7 = 49$$

The age of Sahil 4 years ago was $(49 - 4) = 45$.

Q 146. Ans. C**Explanation:**

Formula to find the angle between hour hand and minute hand is:

$$\text{Angle} = |(11 \times \text{minutes} - 60 \times \text{hours})/2|^\circ$$

Here, minutes = 36, Hours = 2.

$$\text{Angle} = |(11 \times 36 - 60 \times 2)/2|^\circ$$

$$\text{Angle} = |(396 - 120)/2|^\circ$$

$$\text{Angle} = 276/2 = 138^\circ$$

Q 147. Ans. B**Explanation:**

- Conclusion I does not follow:
- The statement only provides the result (no student failed), but it does not provide any reason for it. We cannot conclude that the exam was "very easy." It may also be possible that students were well-prepared or teaching was effective. Hence, this is an assumption, not a logical conclusion.

Q 148. Ans. C**Explanation:**

$$\frac{-7}{9} = -0.777$$

$$\frac{-5}{6} = -0.722...$$

$$\frac{-5}{6} = -0.833...$$

$$\frac{-15}{24} = -0.625$$

Among these, the smallest number is $-0.833...$

Therefore, the smallest rational number = $-5/6$.

Q 149. Ans. B**Explanation:**

Standard Discount = 20%

Discount given to the friend = 25%

Difference in discount = $25\% - 20\% = 5\%$

Loss incurred due to this additional 5% = ₹150

1% = ₹30

Marked Price of the mobile phone = 100%

100% = ₹30 × 100 = ₹3000

Marked Price of the mobile phone = ₹3,000

Q 150. Ans. A**Explanation:**

Let B = 100

Then A = $100 + 25 = 125$

Now,

$$A - B = 125 - 100 = 25$$

C is 200% more than (A - B).

$$\Rightarrow C = 25 + 200\% \text{ of } 25$$

$$\Rightarrow C = 25 + 50 = 75$$

Now, B = 100 and C = 75

Difference = $100 - 75 = 25$

Percentage decrease = $25/100 \times 100 = 25\%$

Therefore, C is 25% less than B.

73rd BPSC Foundation Batch

Prelims + Mains + Interview

BATCH 9

हिंदी माध्यम

📍 Musallahpur

ONLINE & OFFLINE

STARTS FROM 9th APRIL 2026

📍 Boring Road

ONLINE & OFFLINE

STARTS FROM 21st APRIL 2026

BATCH 6

Bilingual

BATCH 1

English Medium

📍 Boring Road

ONLINE & OFFLINE

STARTS FROM 21st APRIL 2026



Patna

Musallahpur

KGS Campus,
Near Sai Mandir,
Musallahpur Hat,
Patna, Bihar-800006

Boring Road

2nd Floor, Kumar
Tower, Boring Road
Chauraha, Patna,
Bihar-800001

Naya Tola

KP Yadav
Compound Kazipur,
Naya Tola, Patna,
Bihar-800004